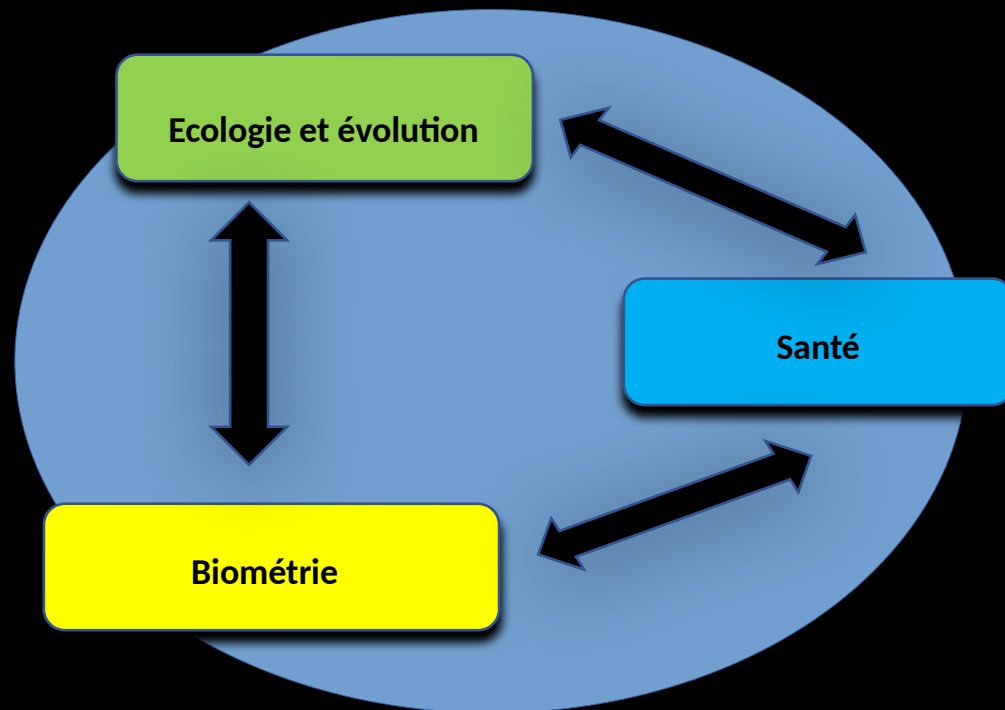


Analyse des structures spatiales multivariées - le package adespatial -

Stéphane Dray

INED - 2026



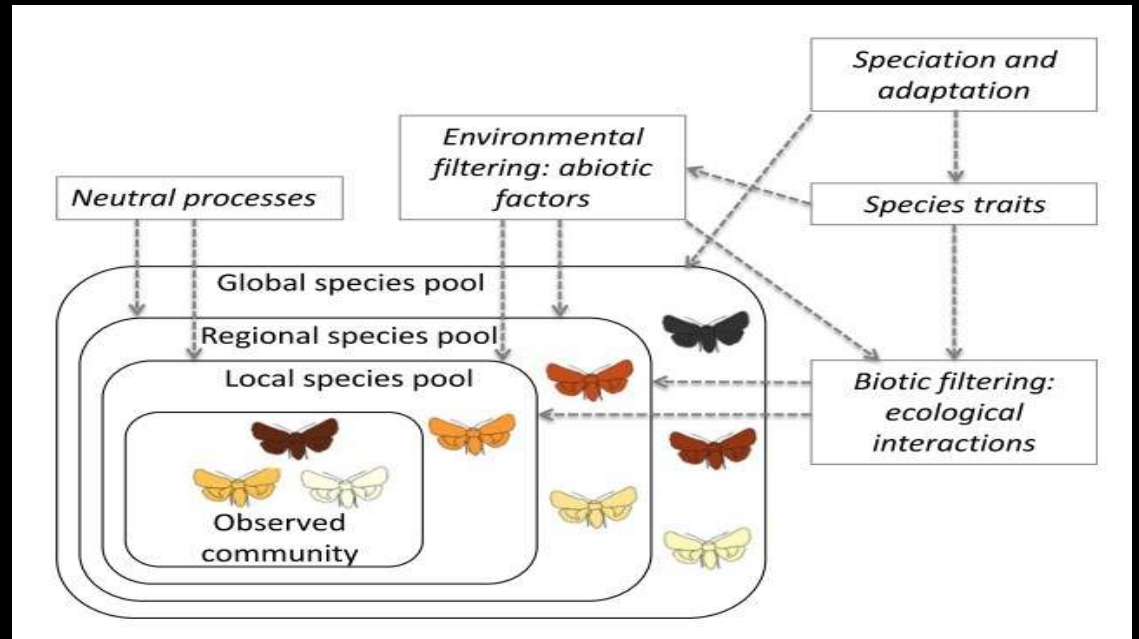
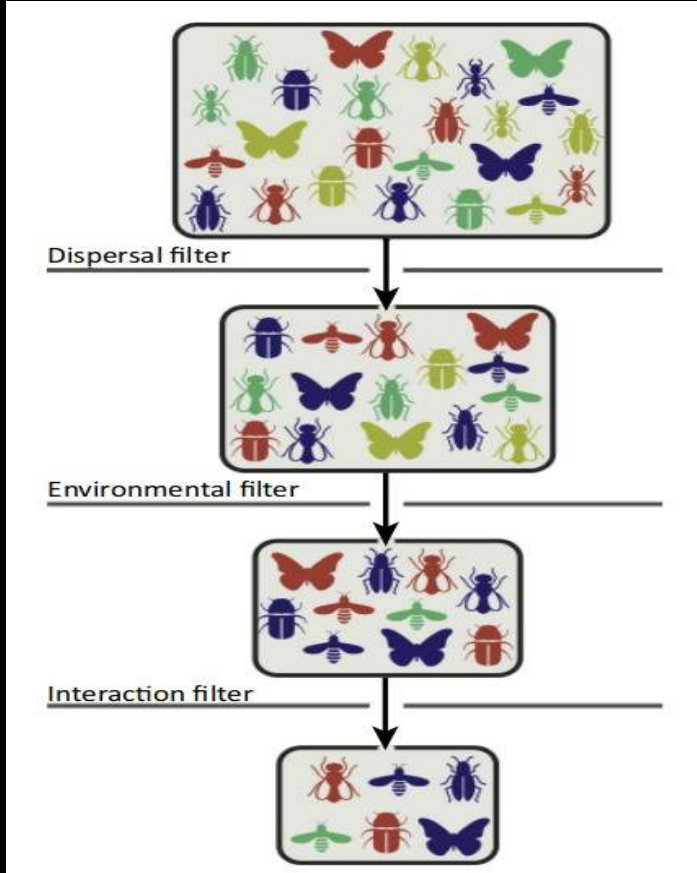
Communautés écologiques



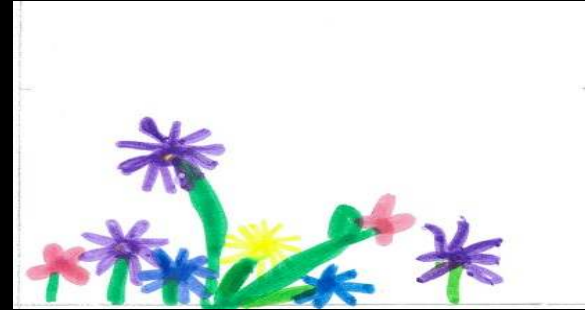
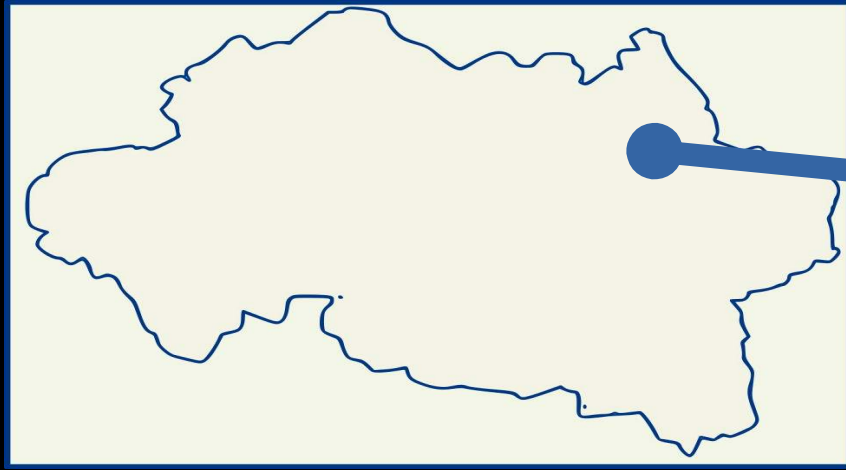
Écologie des communautés

- Décrire les communautés (assemblage d'espèces)
- Étudier les variations de composition (dans l'espace et le temps)
- Identifier les facteurs/processus sous-jacents

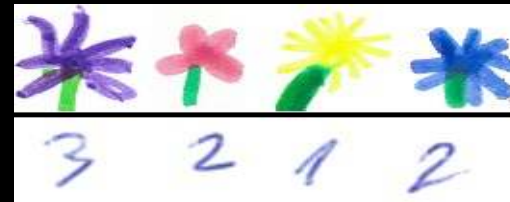
De nombreux processus



Une communauté

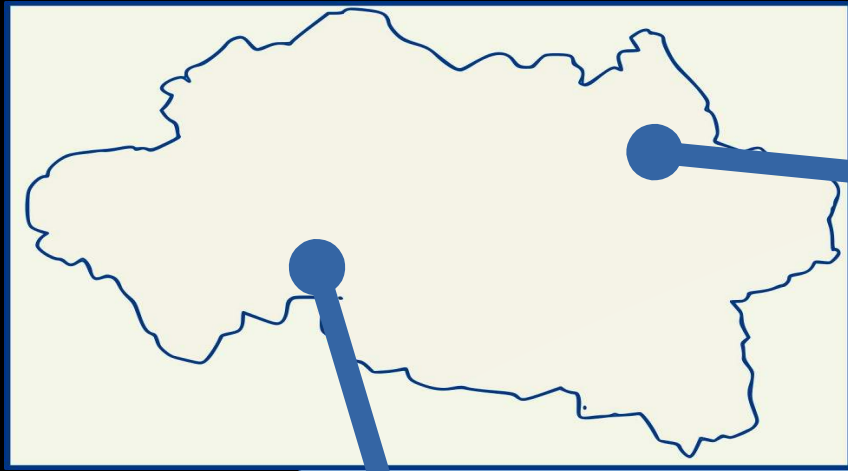


Un vecteur :



Diversité α

Deux communautés



Diversité α_1



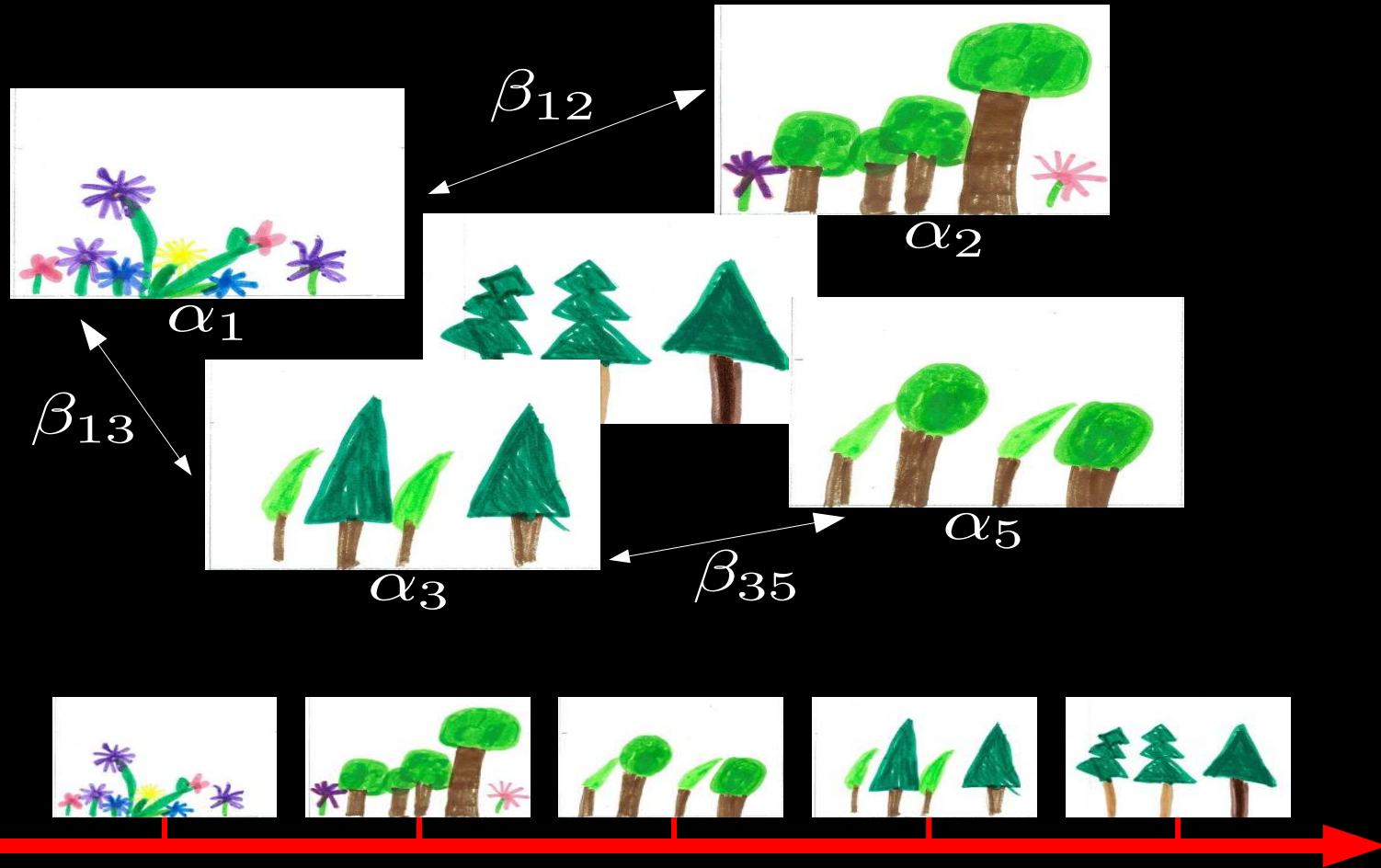
Diversité α_2

Diversité β_{12}

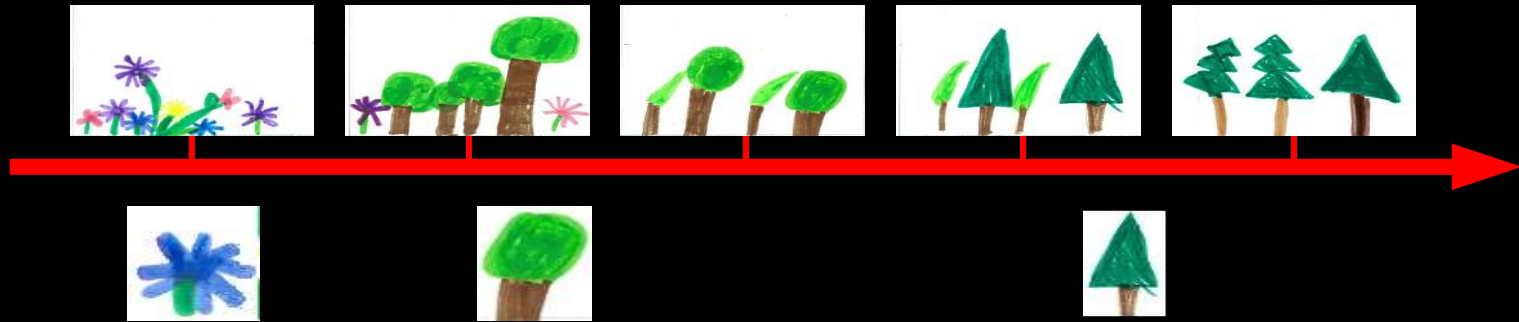
Diversité γ

0	3	2	1	2
4	1	1	0	0

Ordination des communautés

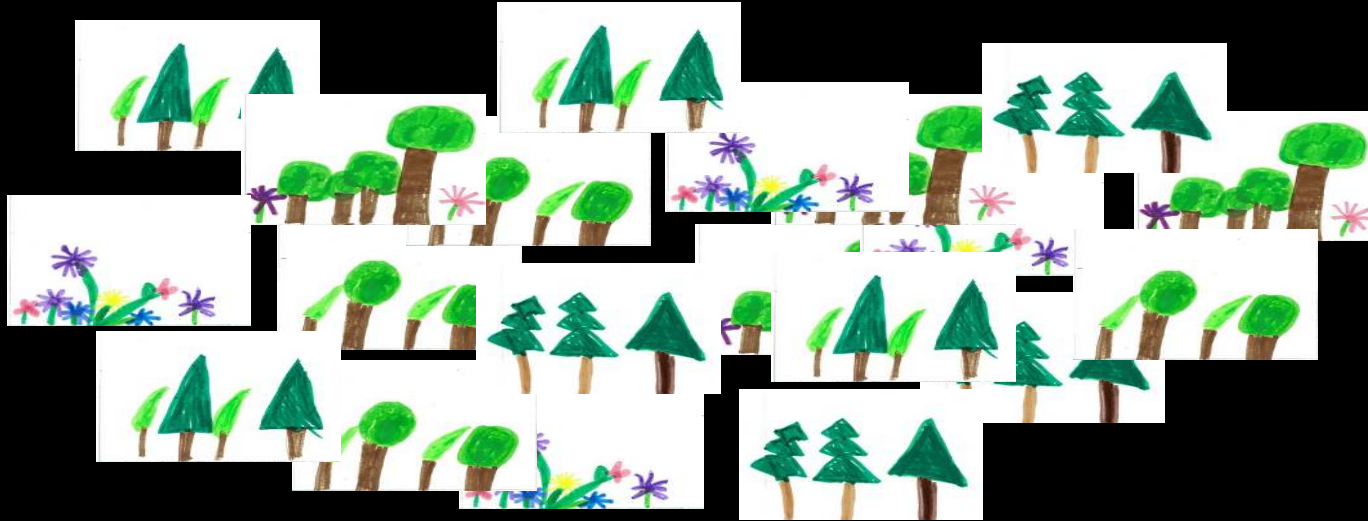


Ordination simultanée

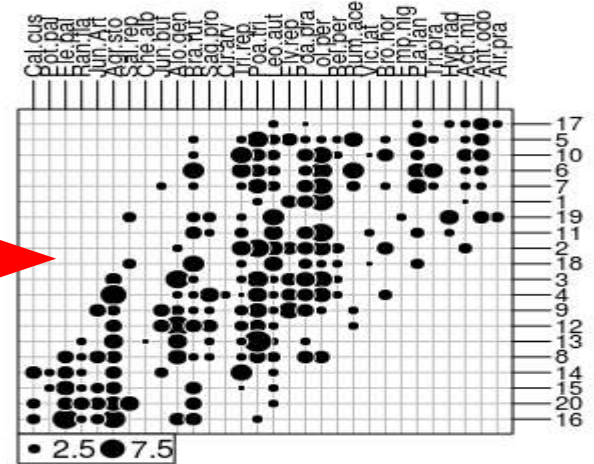
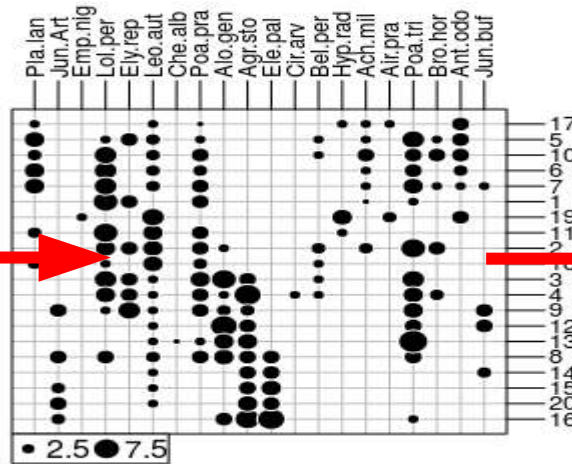
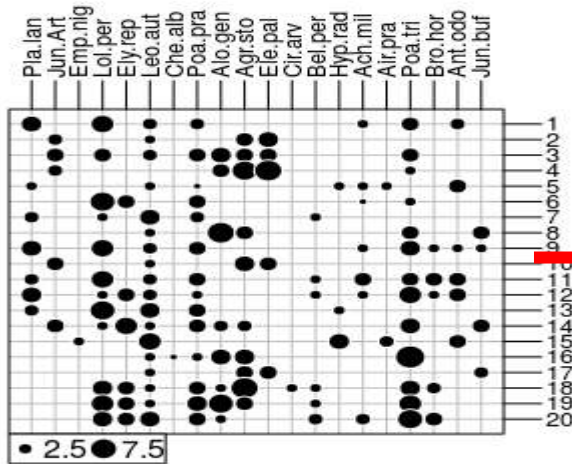


Moyenne pondérée
(*weighted averaging*)

Un casse-tête



Ordonner un tableau



Besoin de méthodes objectives permettant d'ordonner les sites et les espèces et d'identifier les principaux axes de variation de diversité.

D'une question écologique à une problématique méthodologique

OBJECTIVE METHODS FOR THE CLASSIFICATION OF VEGETATION

III. AN ESSAY IN THE USE OF FACTOR ANALYSIS*

By D. W. GOODALL†

(Manuscript received April 5, 1954)

AN ORDINATION STUDY OF A CHALK GRASSLAND COMMUNITY

By M. P. AUSTIN*

GRADIENT ANALYSIS OF VEGETATION*

By

R. H. WHITTAKER

AN ORDINATION OF THE UPLAND FOREST COMMUNITIES OF SOUTHERN WISCONSIN*

J. ROGER BRAY† AND J. T. CURTIS

*Department of Botany, University of Minnesota, Minneapolis, Minnesota
Department of Botany, University of Wisconsin, Madison, Wisconsin*

GEOMETRIC MODELS IN ECOLOGY

I. THE THEORY AND APPLICATION OF SOME ORDINATION METHODS

By L. ORLOCI

THE APPLICATION OF QUANTITATIVE METHODS TO VEGETATION SURVEY

II. SOME METHODOLOGICAL PROBLEMS OF DATA
FROM RAIN FOREST

By M. P. AUSTIN* AND P. GREIG-SMITH

PRINCIPAL COMPONENT ORDINATION AND SIMULATED VEGETATIONAL DATA¹

I. NOY-MEIR

*Department of Biogeography and Geomorphology, Research School of Pacific Studies
Australian National University, Canberra, Australia*

AND

M. P. AUSTIN

Division of Land Research, CSIRO, Canberra, Australia

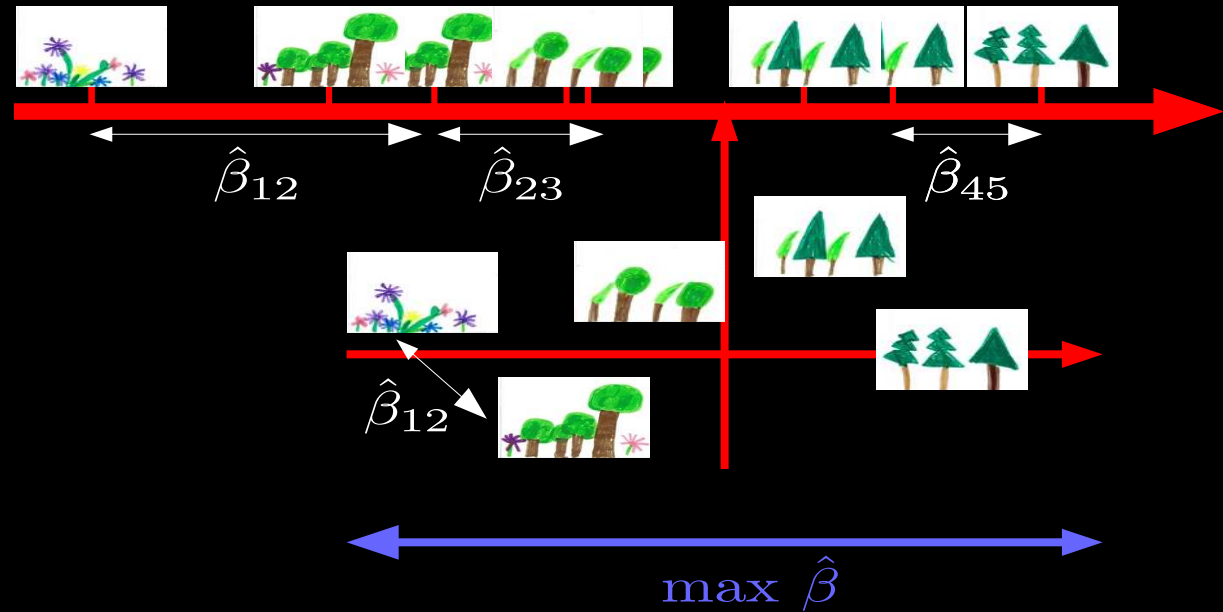
GEOMETRIC MODELS IN ECOLOGY

II. AN EVALUATION OF SOME ORDINATION TECHNIQUES

By M. P. AUSTIN AND L. ORLOCI*

L'ordination des tableaux écologiques

- Scoring
- Multi-dimensionnel
- Maximise la diversité



Analyses multivariées

OBJECTIVE METHODS FOR THE CLASSIFICATION OF
VEGETATION

III. AN ESSAY IN THE USE OF FACTOR ANALYSIS*

By D. W. GOODALL†

(Manuscript received April 5, 1954)

ACP (1954)

AN ORDINATION OF THE UPLAND FOREST COMMUNITIES OF
SOUTHERN WISCONSIN*

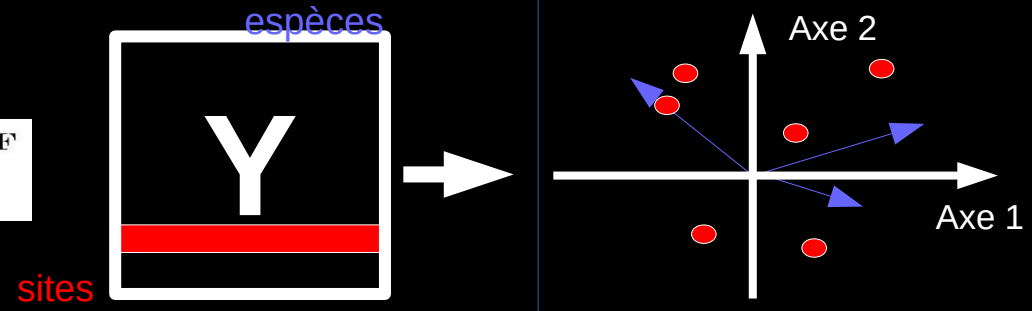
J. ROGER BRAY† AND J. T. CURTIS

PCoA (1957)

RECIPROCAL AVERAGING: AN EIGENVECTOR METHOD OF
ORDINATION

By M. O. HILL*

AFC (1973)



Réconcilier statistiques et écologie

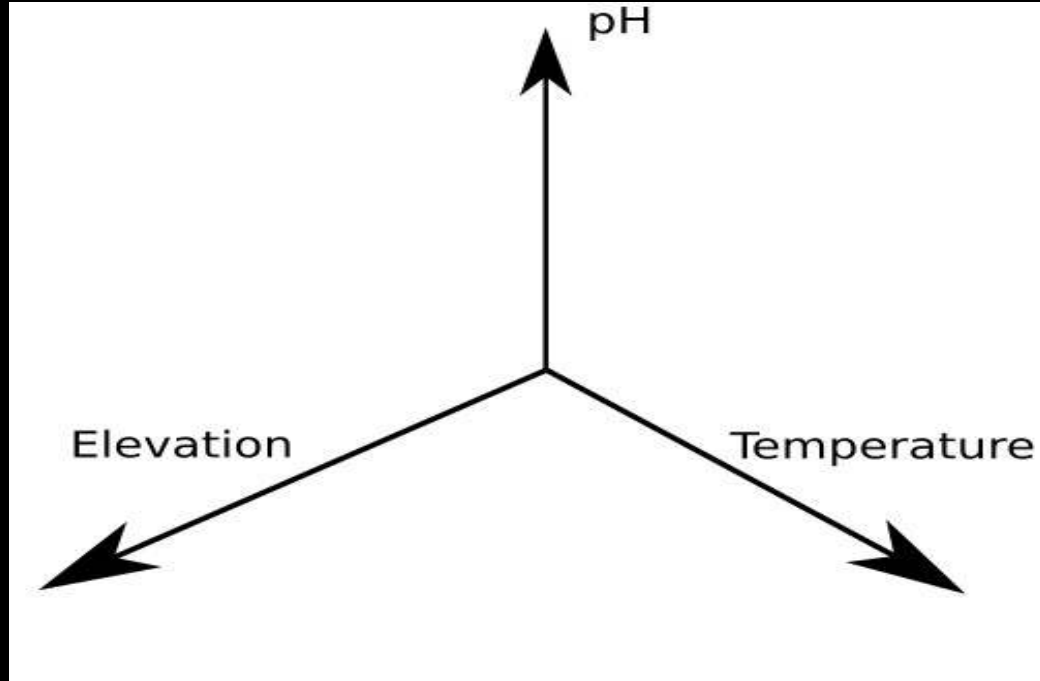
« A major purpose is **interpretation** [...] *and*
not simply the representation of numerical
relationships among samples or species in a
hyperspace with a limited number of axes. »

Patrons ↔ Facteurs / Processus

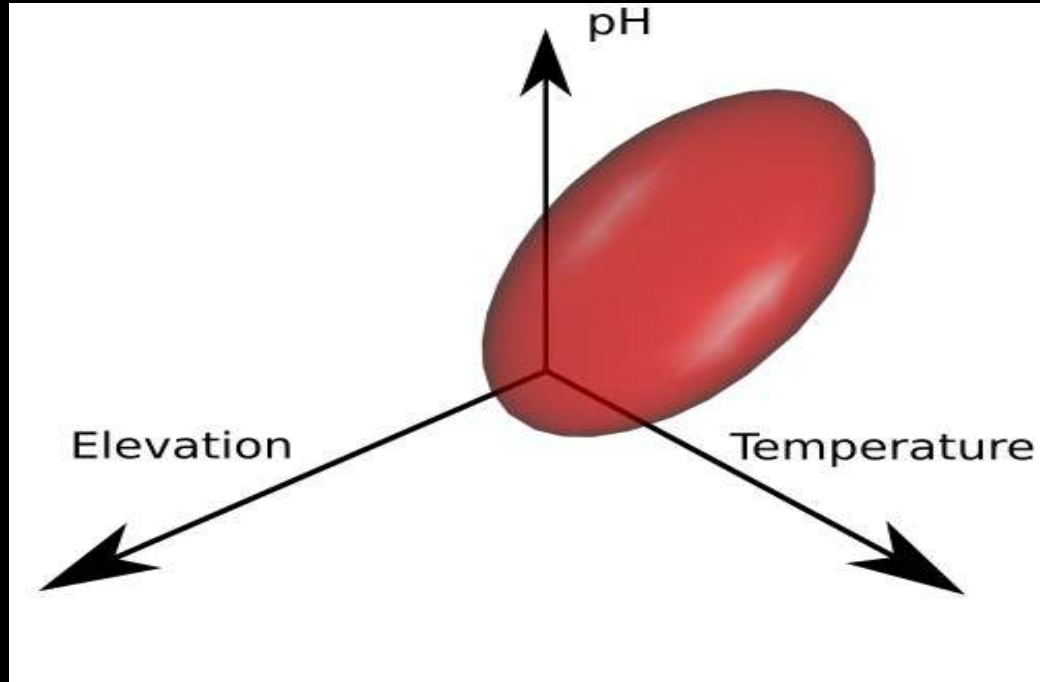
Réconcilier statistiques et écologie

*« A major purpose is **interpretation of community relationships to environment**, and not simply the representation of numerical relationships among samples or species in a hyperspace with a limited number of axes. »*

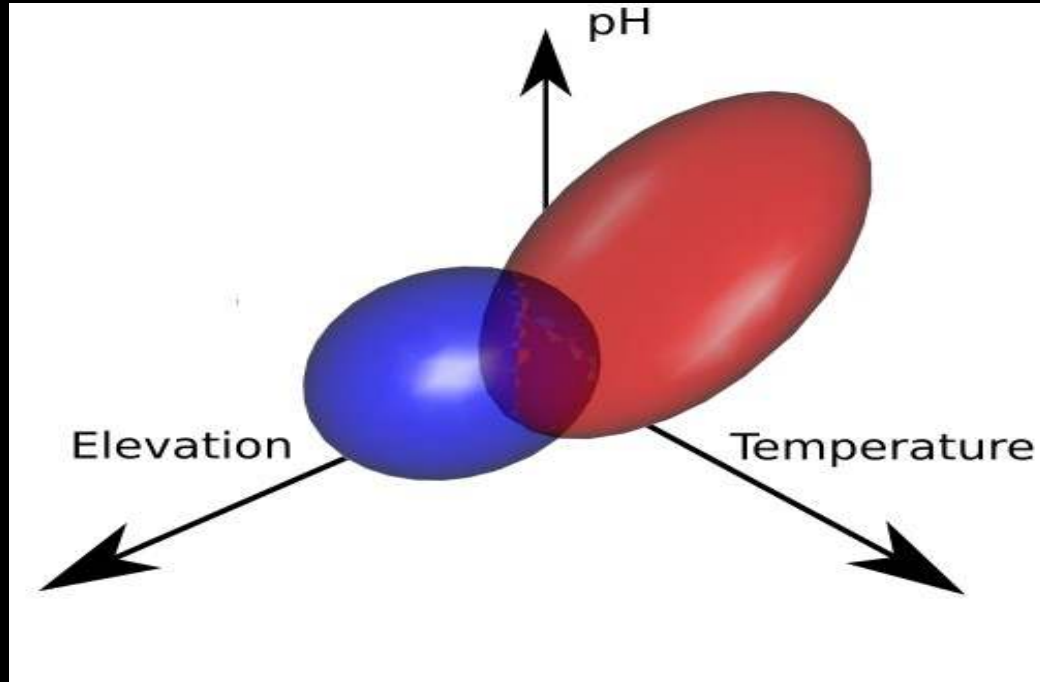
La théorie de la niche



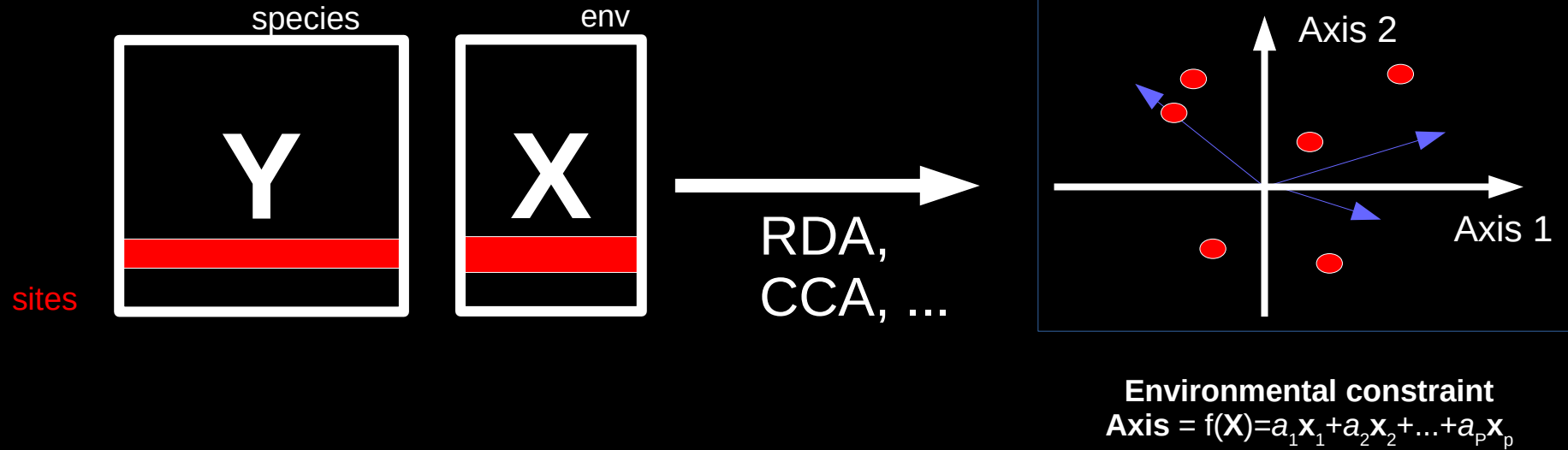
La théorie de la niche



La théorie de la niche



Couplage de tableaux



Ordination directe

- Redundancy Analysis (Rao 1964)
- Canonical Correspondence Analysis (ter Braak 1986)

$$\mathbf{Y} = \mathbf{XB} + \mathbf{R} = \mathbf{F} + \mathbf{R}$$



Composition prédite



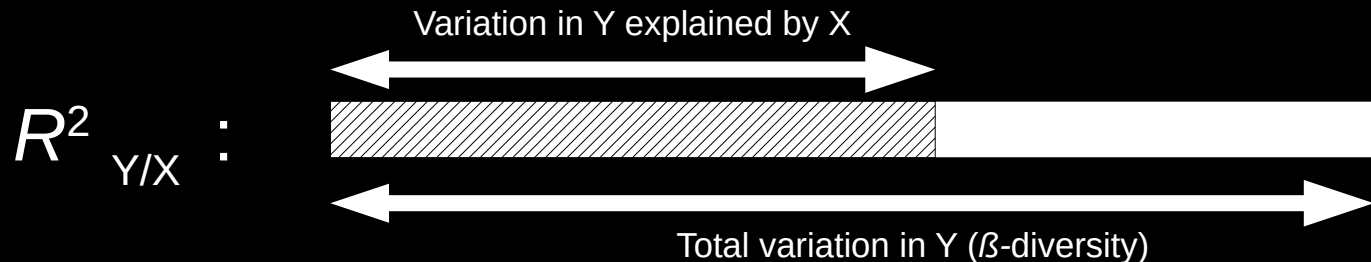
Résidus

Ordination directe

- Redundancy Analysis (Rao 1964)
- Canonical Correspondence Analysis (ter Braak 1986)

$$\mathbf{Y} = \mathbf{XB} + \mathbf{R} = \mathbf{F} + \mathbf{R}$$

- Variation partitioning (Borcard et al 1992)

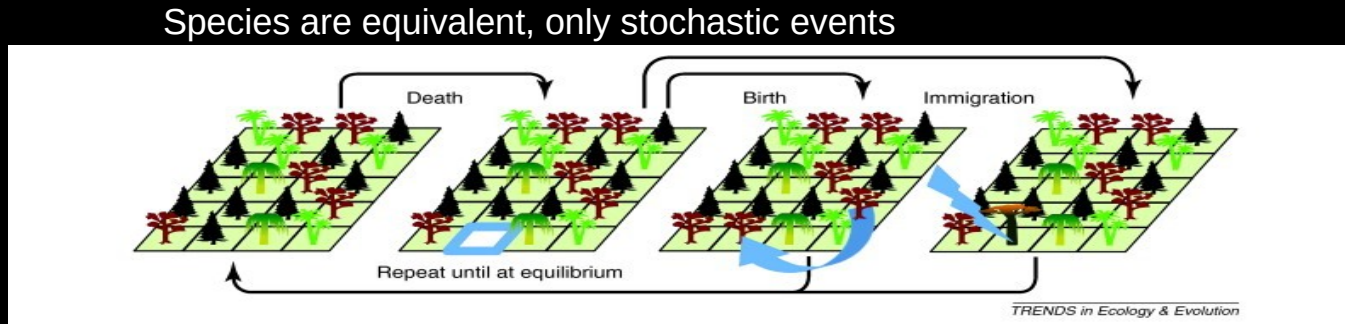


La fin de l'histoire ?

- Variables environnementale non mesurées
- Influence d'autres processus écologiques

La fin de l'histoire ?

- Variables environnementale non mesurées
- Influence d'autres processus écologiques



Hubbell (2001)

Rosindell et al. (2011) TREE

La fin de l'histoire ?

- Variables environnementale non mesurées
- Influence d'autres processus écologiques
- Études expérimentales ou long-terme difficiles

Structures spatiales

- Variables environnementale non mesurées
- Influence d'autres processus écologiques
- Structures spatiales comme signature de variables non mesurées ou de processus inconnus

Ecology, 90(1), 2009, pp. 46–56
© 2009 by the Ecological Society of America

Beyond description: the active and effective way
to infer processes from spatial patterns

ELIOT J. B. MCINTIRE^{1,4} AND ALEX FAJARDO^{2,3}

Analyses Spatiales Multivariées

- Comment intégrer l'information spatiale dans une analyse ?
- Quelles méthodes pour faire parler les structures spatiales multivariées ?
- Quand les structures spatiales posent un problème ?

2008

SEDAR

Spatial Ecological Data Analysis with  Workshop

2012

REVIEWS

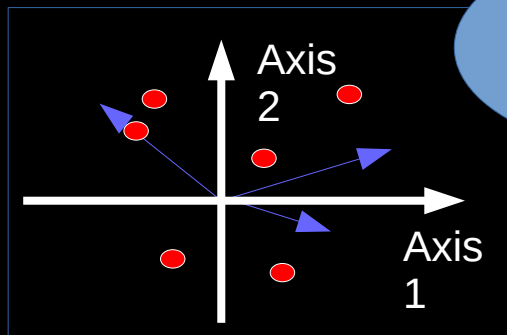
Ecological Monographs, 82(3), 2012, pp. 257–275
© 2012 by the Ecological Society of America

Community ecology in the age of multivariate multiscale spatial analysis

S. DRAY,^{1,16} R. PÉLISSIER,^{2,3} P. COUTERON,² M.-J. FORTIN,⁴ P. LEGENDRE,⁵ P. R. PERES-NETO,⁶ E. BELLIER,^{7,8}
R. BIVAND,⁹ F. G. BLANCHET,¹⁰ M. DE CÁCERES,¹¹ A.-B. DUFOUR,¹ E. HEEGAARD,¹² T. JOMBART,^{1,13} F. MUNOZ,²
J. OKSANEN,¹⁴ J. THIOULOUSE,¹ AND H. H. WAGNER¹⁵

2015

`adespatial` 0.0-1 on CRAN



ade4

adegraphics

vegan

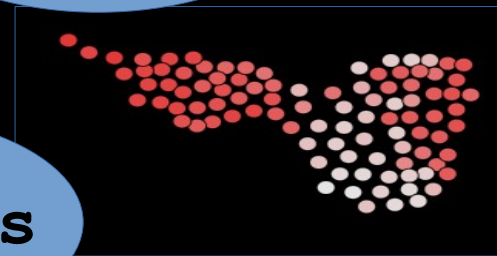
multivarié

adespatial

sp

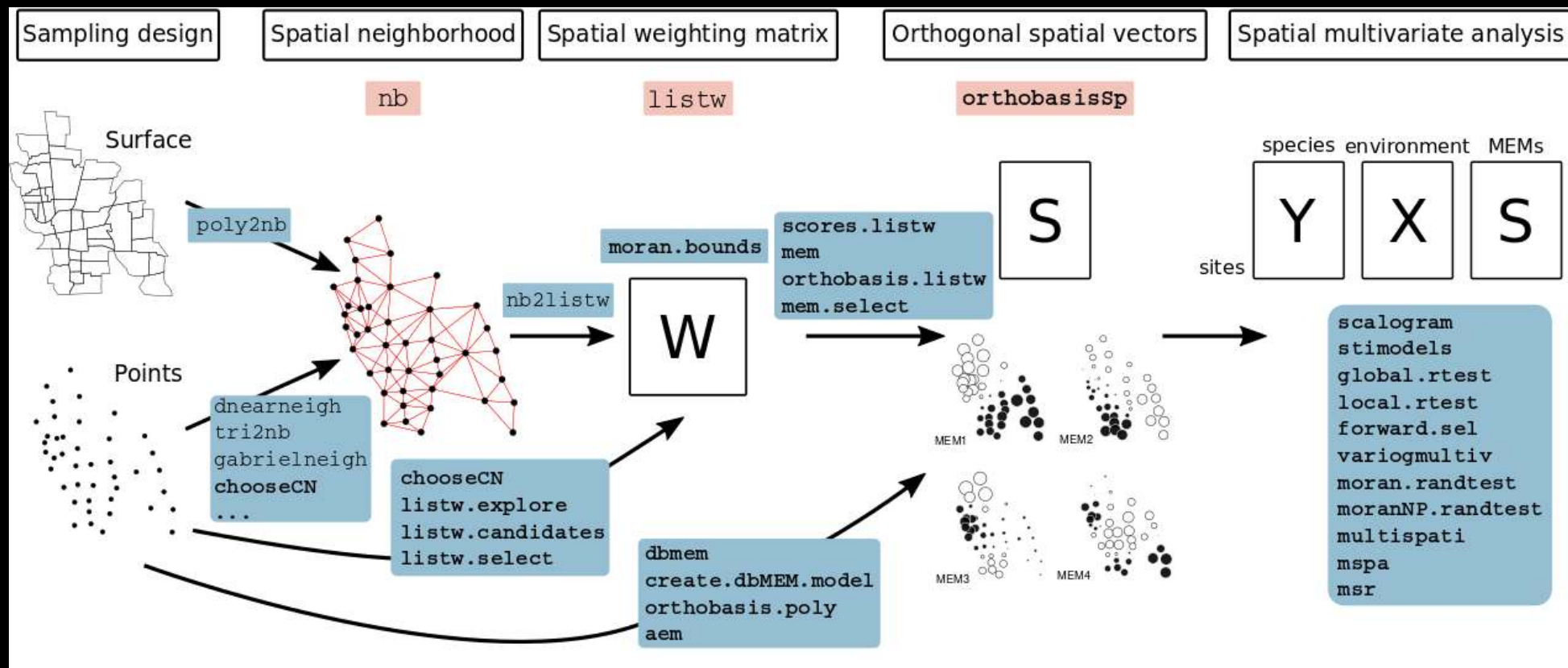
spdep

maptools



spatial

adespatial:



Analyses Spatiales Multivariées

- Comment intégrer l'information spatiale dans une analyse ?
- Quelles méthodes pour faire parler les structures spatiales multivariées ?
- Quand les structures spatiales posent un problème ?

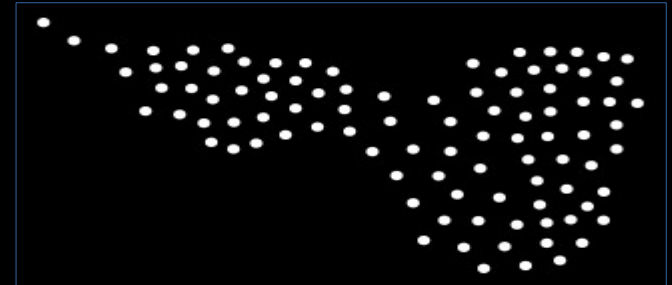
Comment intégrer l'information spatiale?

- Cartographie (Goodall, 1954)
- Distances (Mantel, 1967)
- Poids spatiaux (Moran, 1954)
- Prédicteurs spatiaux (variables):
 - Polynômes des coordonnées (Gittins, 1968)
 - PCNM (Borcard & Legendre, 2002)
 - MEM (Dray et al, 2006)
 - AEM (Blanchet et al, 2008)

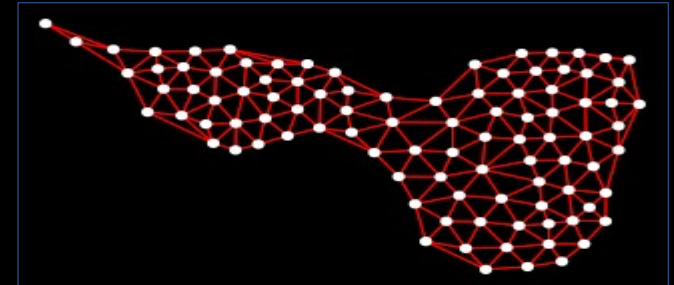
Comment intégrer l'information spatiale?

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 - PCNM (Borcard & Legendre, 2002)
 - MEM (Dray et al, 2006)
 - AEM (Blanchet et al, 2008)

Espace comme graphe (matrice de poids spatiaux)



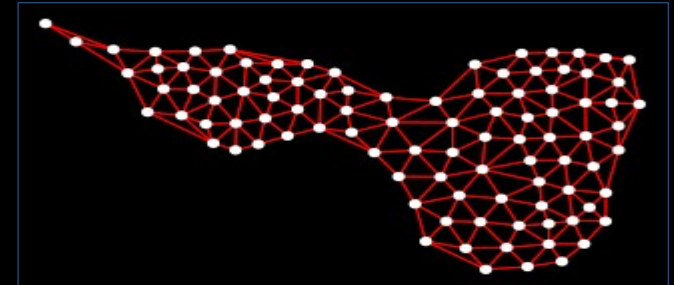
Espace comme graphe (matrice de poids spatiaux)



Espace comme graphe (matrice de poids spatiaux)



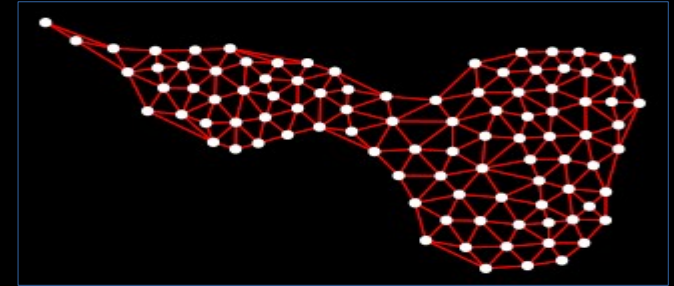
$$\mathbf{W} = \begin{bmatrix} 0 & 1 & \dots & 0 \\ 1 & 0 & \dots & 1 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 1 & \dots & 0 \end{bmatrix}$$



Espace comme graphe (matrice de poids spatiaux)

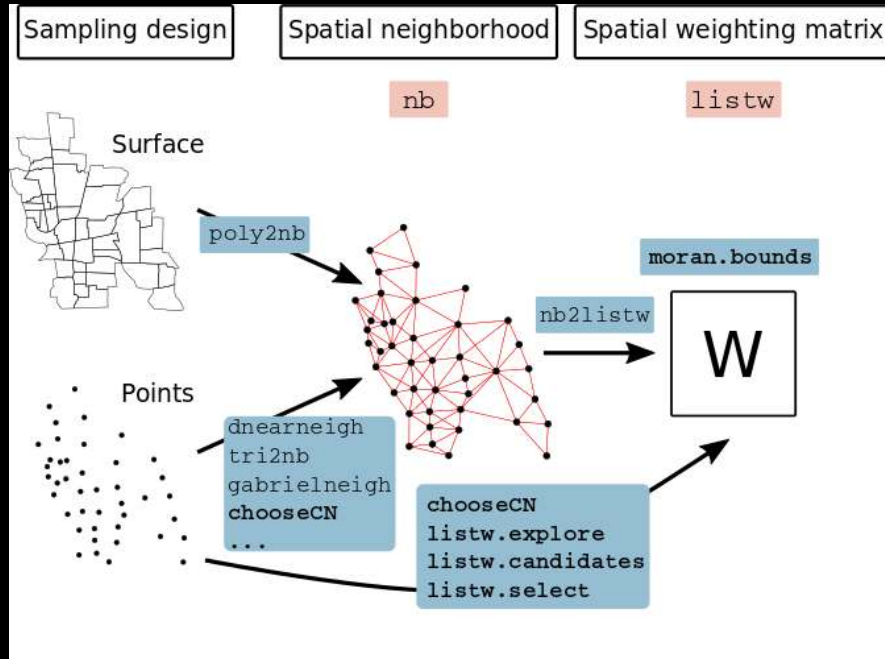


$$\mathbf{W} = \begin{bmatrix} 0 & 1 & \dots & 0 \\ 1 & 0 & \dots & 1 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 1 & \dots & 0 \end{bmatrix}$$



$$MC(\mathbf{x}) = \frac{n}{\mathbf{1}_n^\top \mathbf{W} \mathbf{1}_n} \frac{\mathbf{x}^\top \mathbf{H} \mathbf{W} \mathbf{H} \mathbf{z}}{\mathbf{z}^\top \mathbf{H} \mathbf{H} \mathbf{z}} = \frac{n \sum_{i,j} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i,j} w_{ij} \sum_i (x_i - \bar{x})^2}$$

De l'échantillonnage aux poids spatiaux



Disentangling good from bad practices in the selection of spatial or phylogenetic eigenvectors

David Bauman, Thomas Drouet, Stéphane Dray and Jason Vleminckx

Optimizing the choice of a spatial weighting matrix in eigenvector-based methods

DAVID BAUMAN,^{1,4} THOMAS DROUET,¹ MARIE-JOSÉE FORTIN,² AND STÉPHANE DRAY³

listw.explore

Generate R code to create a spatial weighting matrix

nb options

Sp object or coordinates:

xyir

Graph type:

Gabriel

listw options

Standardization style:

W

General weights:

1 - d/max(d)

R code (copy & paste in the R console):

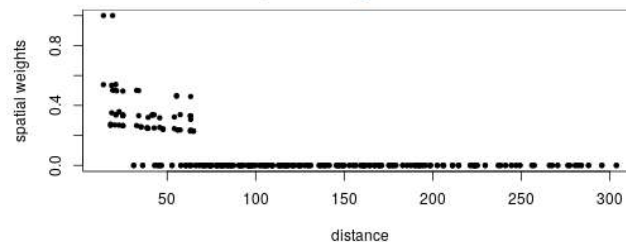
```
library(adespatial);library(sp);library(spdep)
nb <- chooseCN(coordinates(xyir), type = 2, plot.nb = FALSE)
distnb <- nbdists(nb, xyir)
fdist <- lapply(distnb, function(x) 1 - x/max(dist(xyir)))
lw <- nb2listw(nb, style = 'W', glist = fdist, zero.policy =
```

☒ Display summary

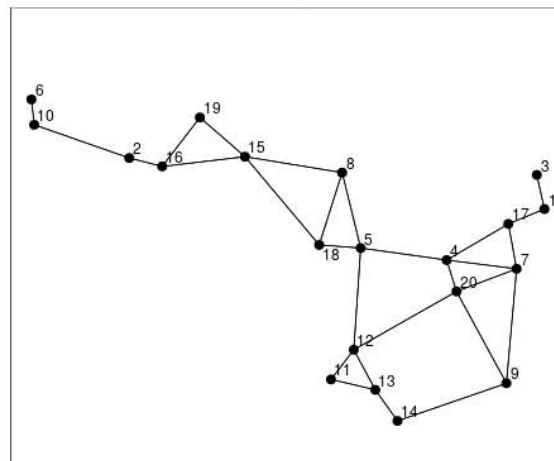
Neighbour list object:
Number of regions: 20
Number of nonzero links: 56
Percentage nonzero weights: 14
Average number of links: 2.8
Link number distribution:

1 2 3 4

Spatial weights



Connectivity



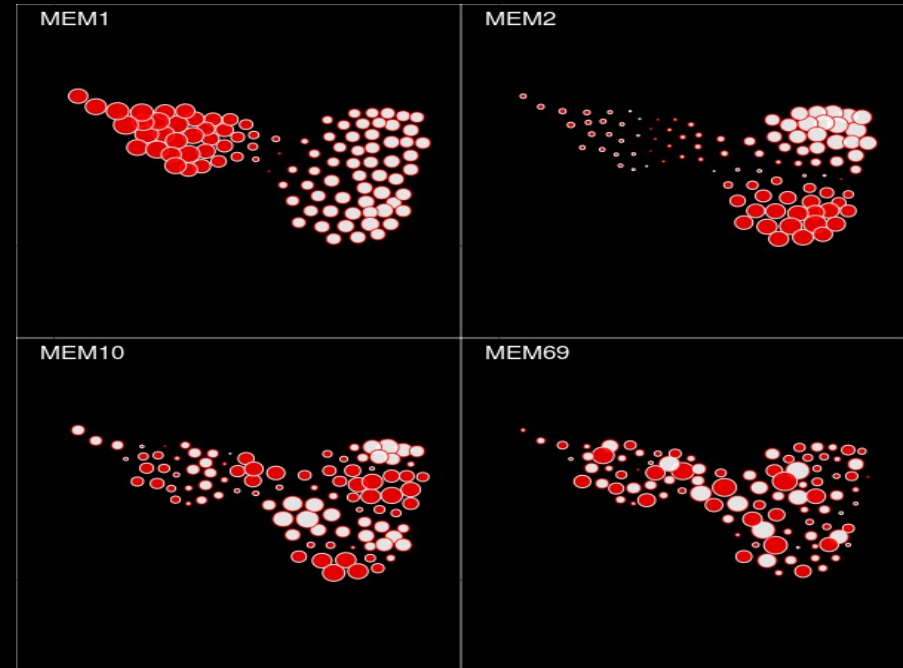
Moran's Eigenvector Maps

$$\mathbf{H}\mathbf{W}\mathbf{H}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}$$

Eigenvectors are orthogonal and maximize Moran's coefficient :

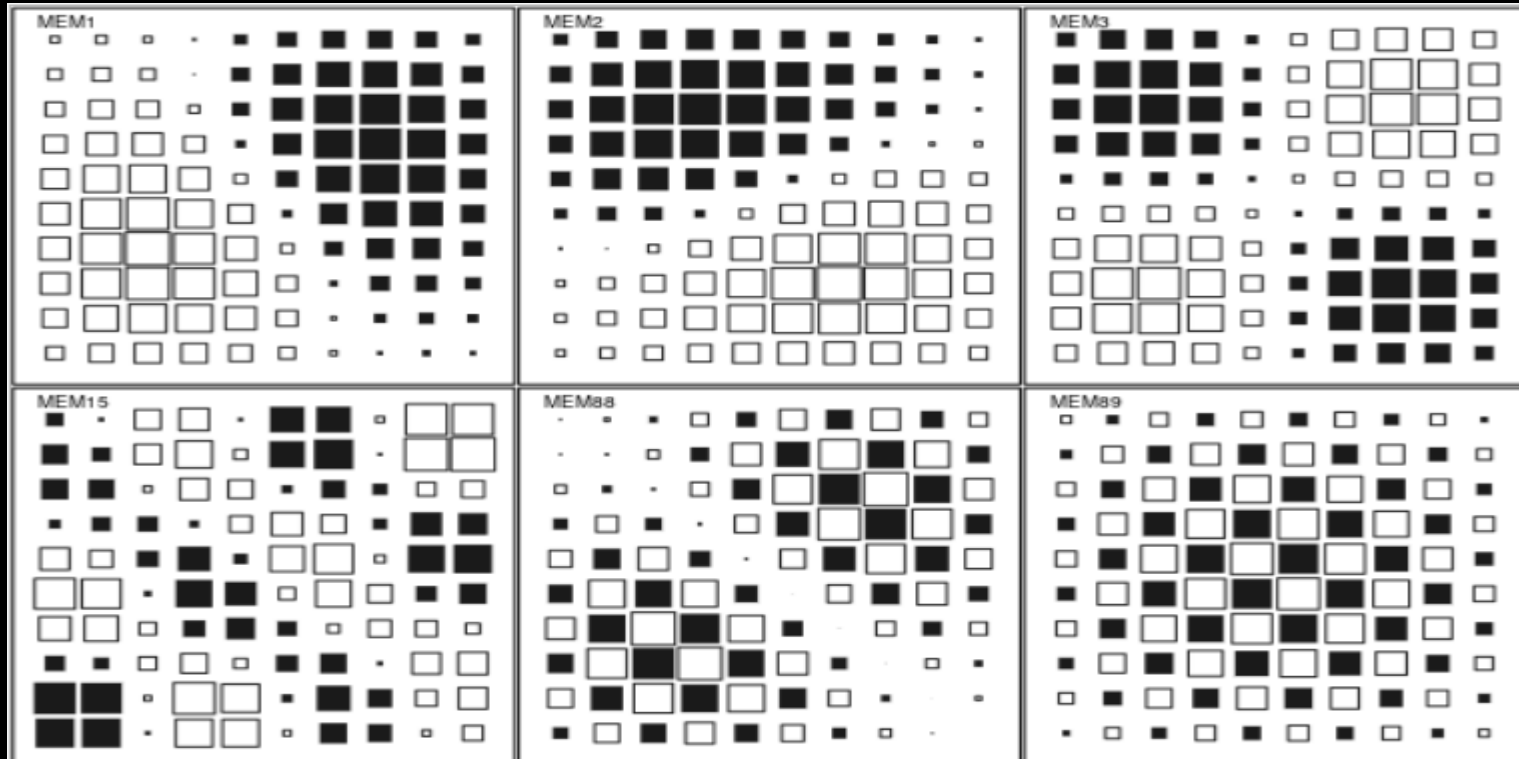
$$MC(\mathbf{v}_i) = \lambda_i$$

```
mymem <- mem(mylistw)  
plot(mymem[,c(1,2,10,69)], xy)
```

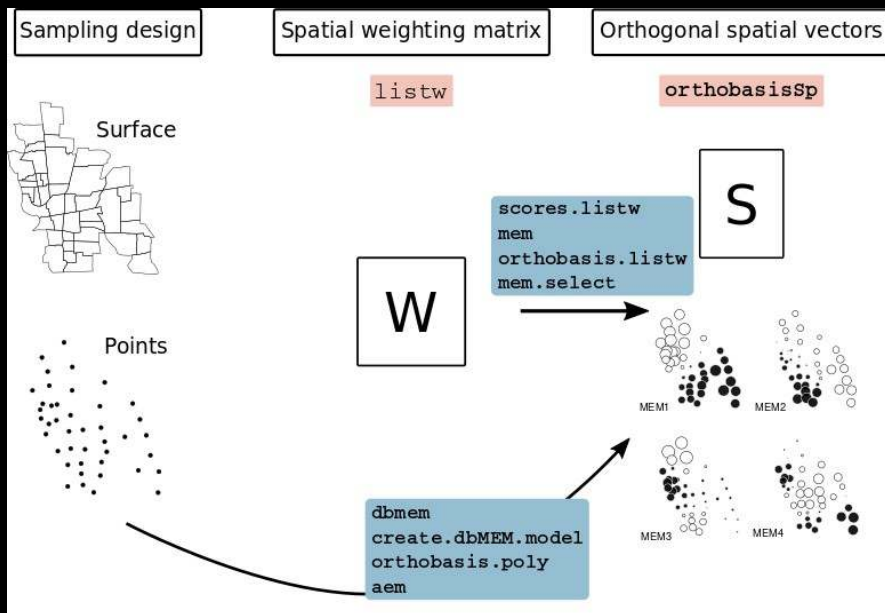


Moran's Eigenvector Maps

Échantillonnage régulier 2D



Des matrices de poids aux prédicteurs spatiaux



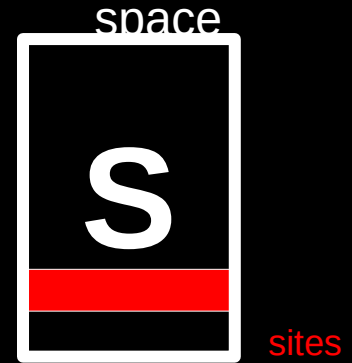
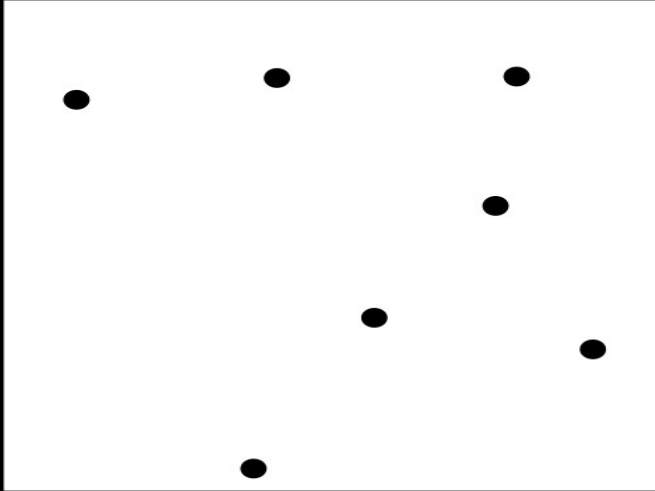
orthobasisSp

- `plot`, `summary`, `print` methods
- inherits from `data.frame`

Analyses Spatiales Multivariées

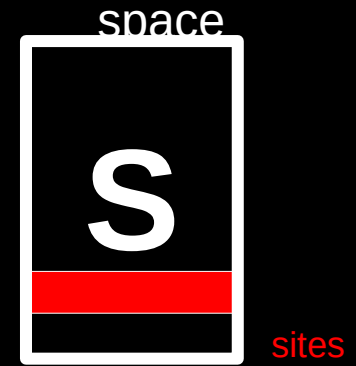
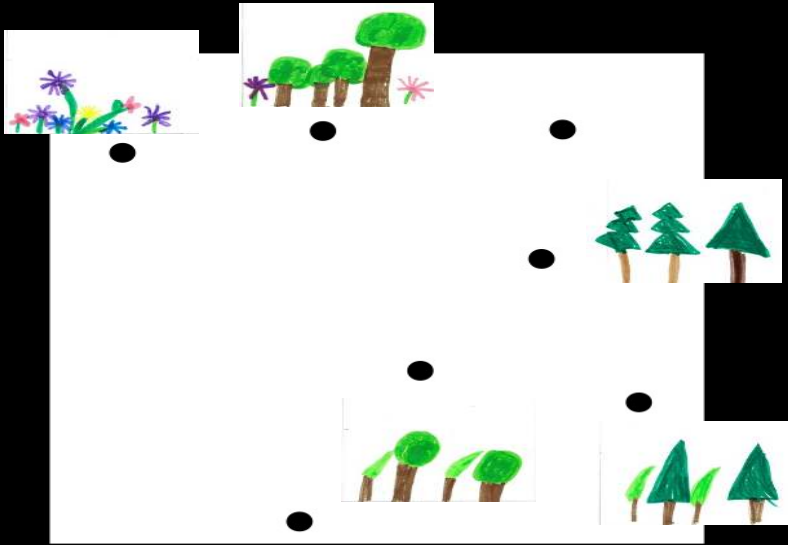
- Comment intégrer l'information spatiale dans une analyse ?
- Quelles méthodes pour faire parler les structures spatiales multivariées ?
- Quand les structures spatiales posent un problème ?

Information spatiale



- Poids spatiaux
- MEM
- Distances

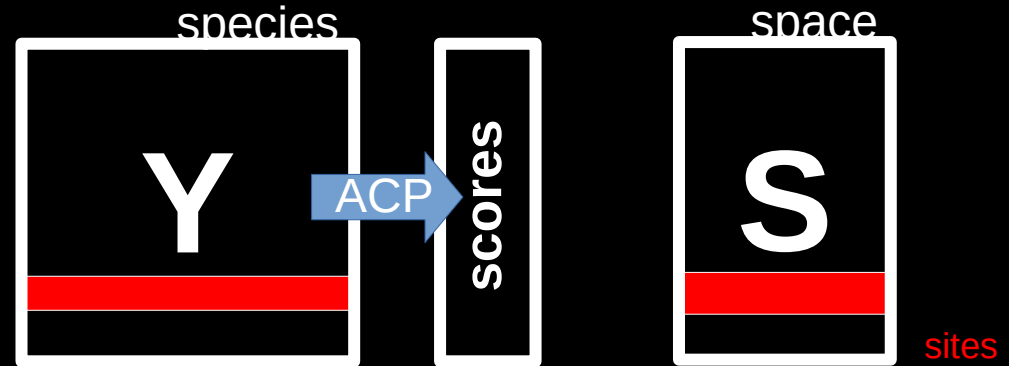
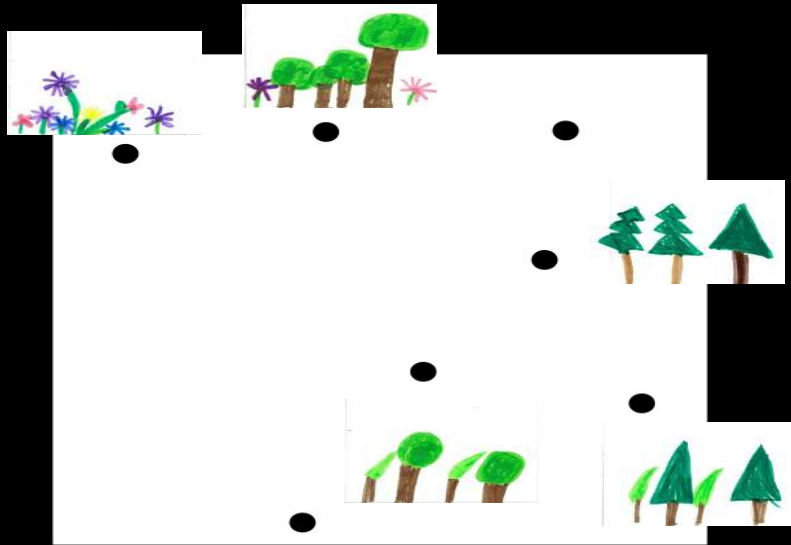
Comment la composition varie dans l'espace ?



- Poids spatiaux
- MEM
- Distances

Comment la composition varie dans l'espace ?

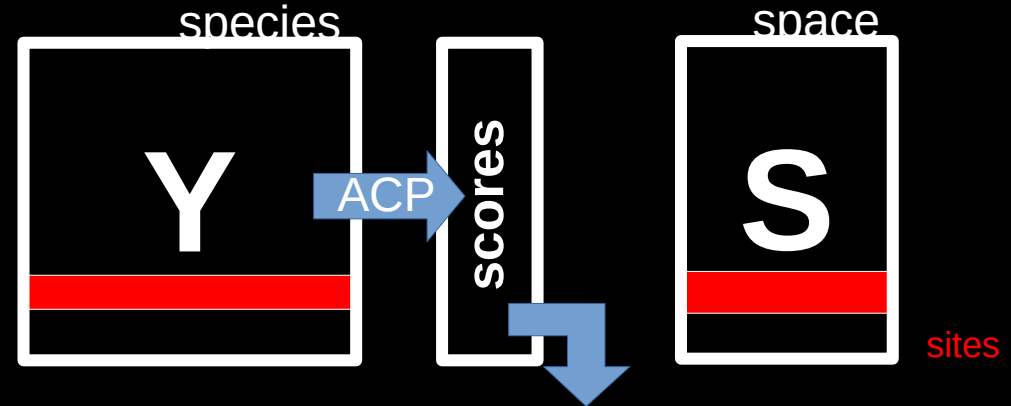
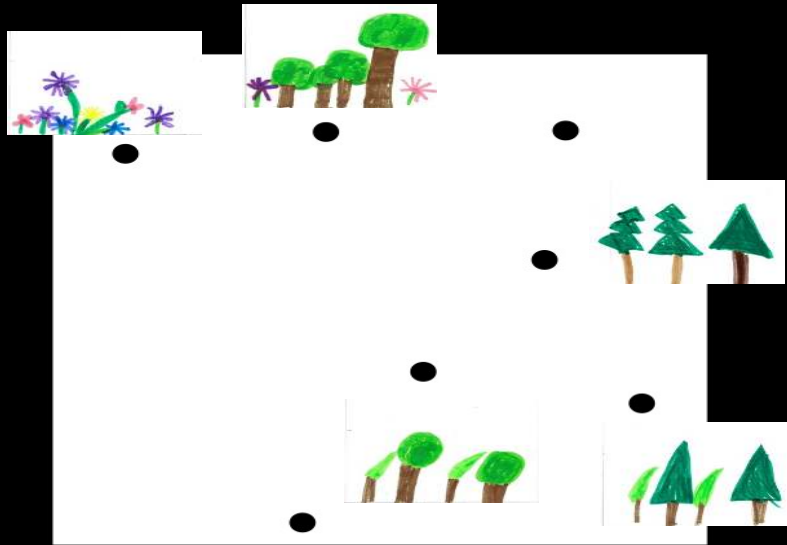
Approche indirecte



- Poids spatiaux
- MEM
- Distances

Comment la composition varie dans l'espace ?

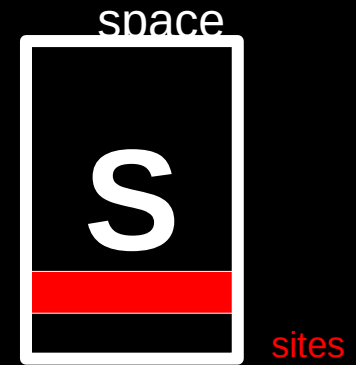
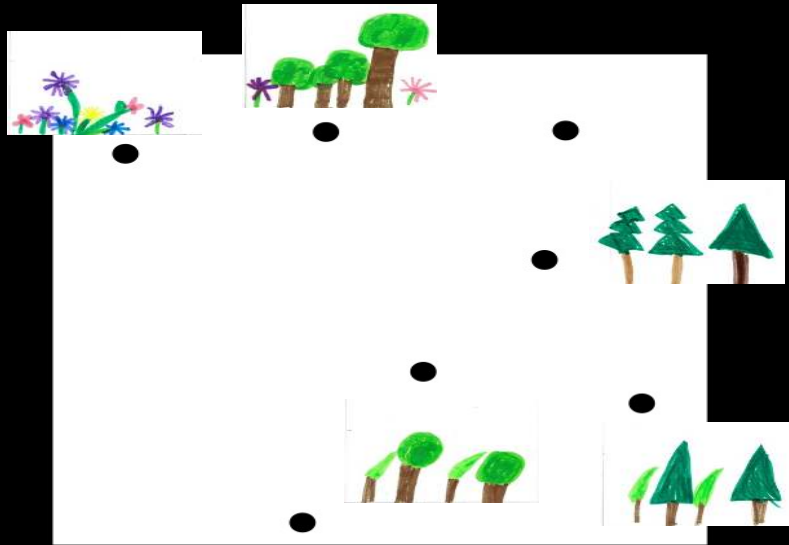
Approche indirecte



Indice de Moran	←	Poids spatiaux
Régression	←	MEM
Test de Mantel	←	Distances

Comment la composition varie dans l'espace ?

Approche directe



Multispati

RDA / MSPA

Test de Mantel



Poids spatiaux

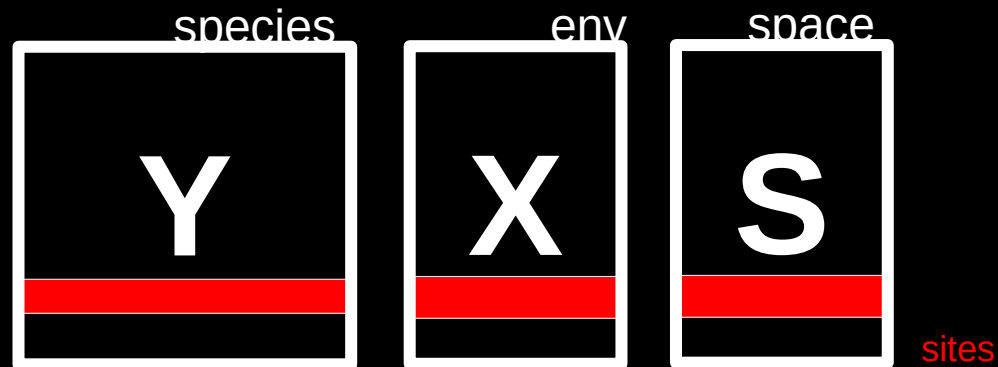
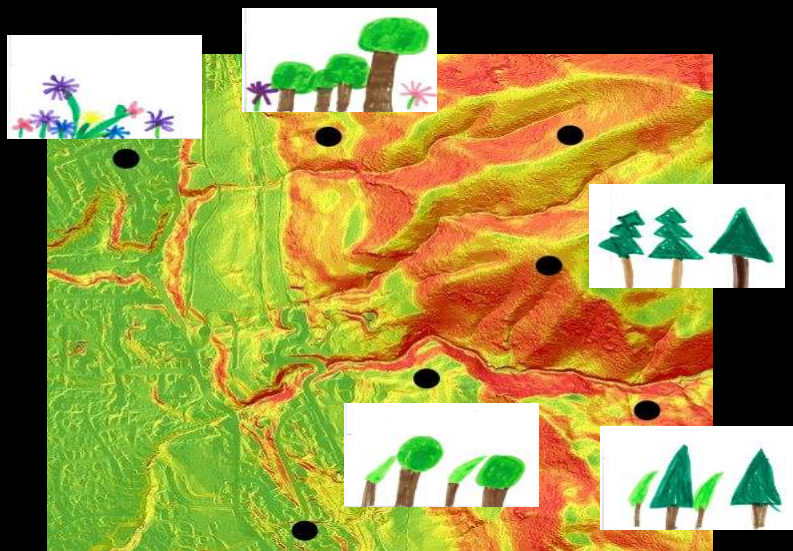


MEM



Distances

Pourquoi la composition varie dans l'espace ?

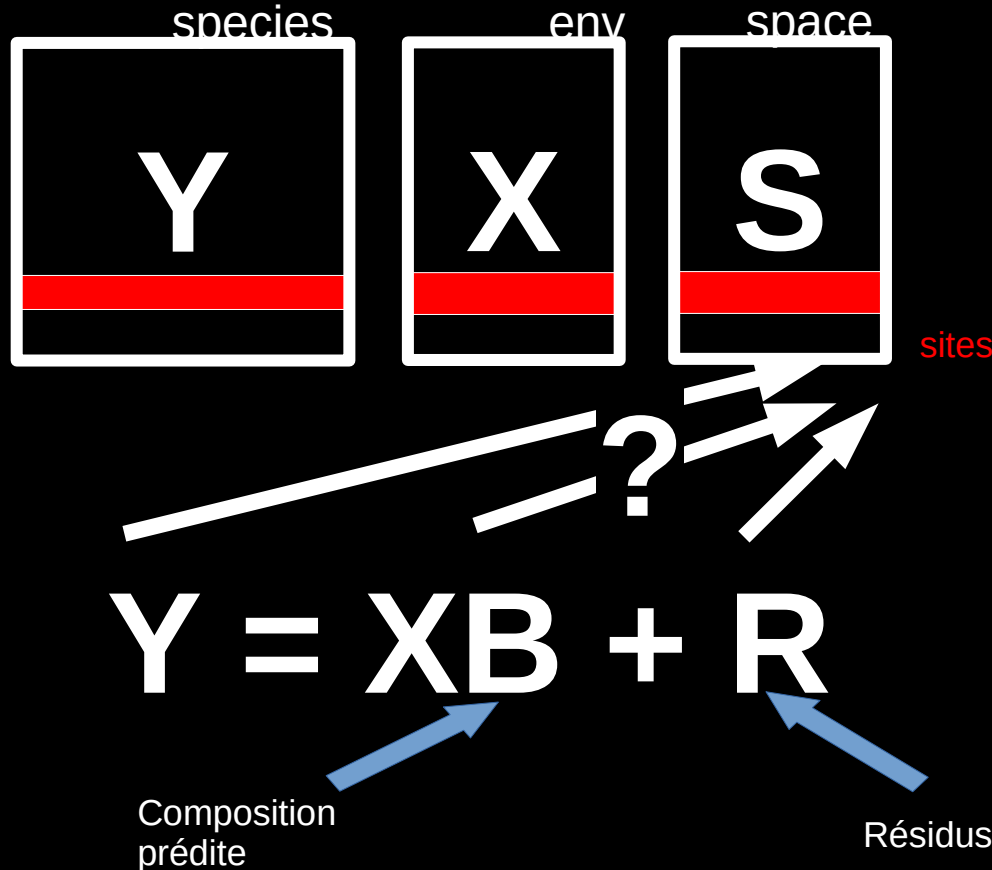


$$Y = XB + R$$

Composition
prédite

Résidus

Pourquoi la composition varie dans l'espace ?



La présence et la nature des structures spatiales observées, prédites et résiduelle peut être étudiée ou prise en compte.

Structures spatiales comme proxy

$$\begin{aligned} Y &= XB + R \\ &= XB_S + XB_{NS} + R_S + R_{NS} \end{aligned}$$

Résidus
(autocorrélés)

Proxy de processus ou
variables non mesurées

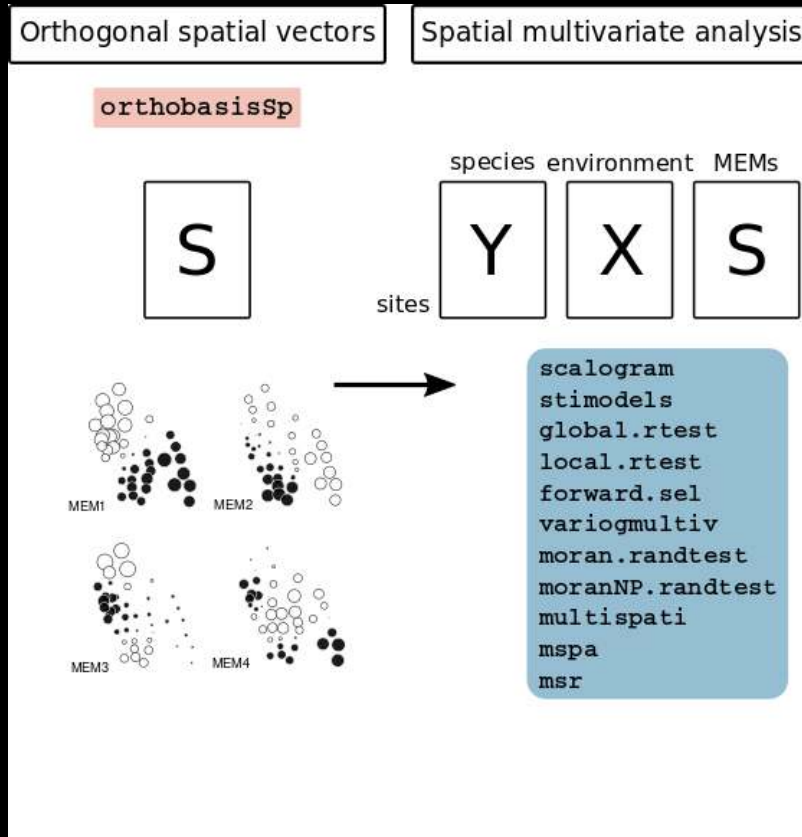
Résidus
indépendants

Structures spatiales comme proxy

$$\begin{aligned} Y &= XB + R \\ &= XB_S + XB_{NS} + R_S + R_{NS} \end{aligned}$$

- Structures spatiales liées à l'environnement ($Y \sim X \leftrightarrow S$) : **XB_S**
- Structures spatiales liées à d'autres facteurs ($Y|X \leftrightarrow S$) : **R_S**
- Quelles échelles ?

Méthodes spatiales multi-échelles et/ou multivariées



Méthodes spatiales multivariées

- Multivariate autocorrelation:

`global.rtest(Y, mylistw)`

Jombart et al (2008) Heredity

- Spatial multivariate analysis:

`multispati(mypca, mylistw)`

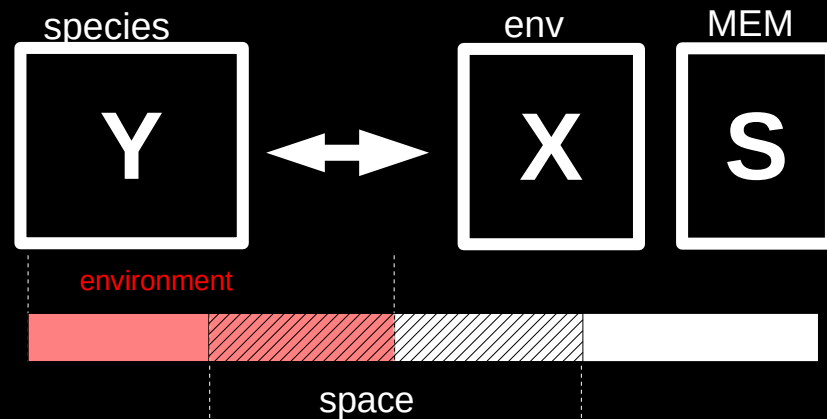
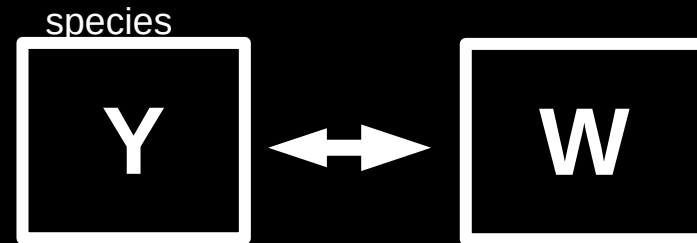
Dray et al (2008) Journal of Vegetation Science

- Constrained ordination/Variation partitioning:

`pcaiv(mypca, mymem)`

`varipart(Y, X, mymem)`

Dray et al (2012) Ecological Monographs



Méthodes multi-échelles

- Moran's Index: `moran.randtest(x, mylistw)`
- Positive/Negative decomposition: `moranNP.randtest(x, mylistw)`

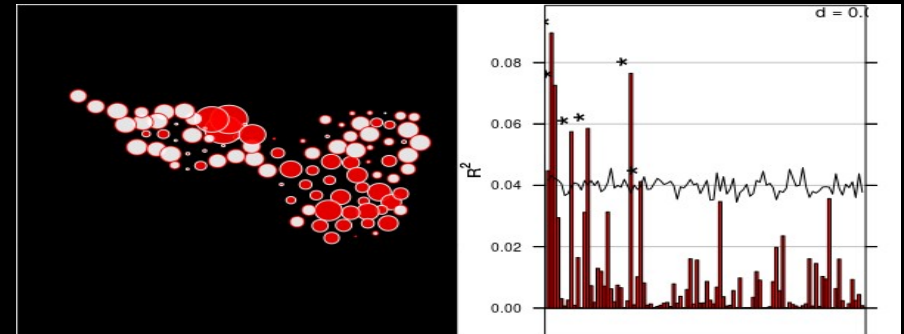
$$MC(\mathbf{x}) = \sum_{i=1}^{n-1} \lambda_i \text{cor}^2(\mathbf{x}, \mathbf{v}_i)$$

Dray (2011) Geographical Analysis

- Multiscale decomposition:

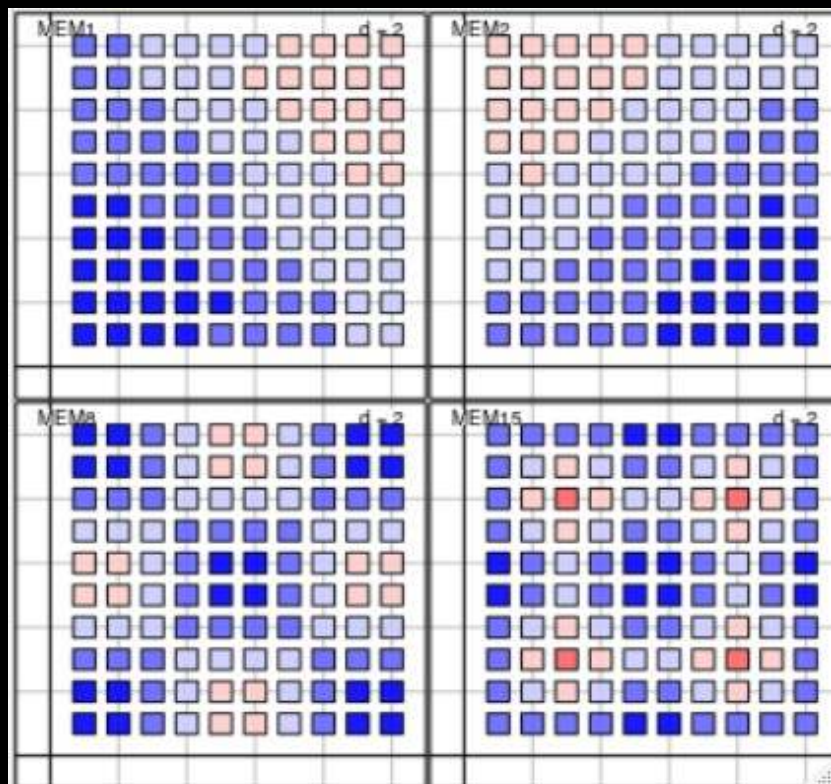
`scalogram(x, mymem)`
`mspa(mypca, mymem)`

$$\sum_{i=1}^{n-1} \text{cor}^2(\mathbf{x}, \mathbf{v}_i) = 1$$

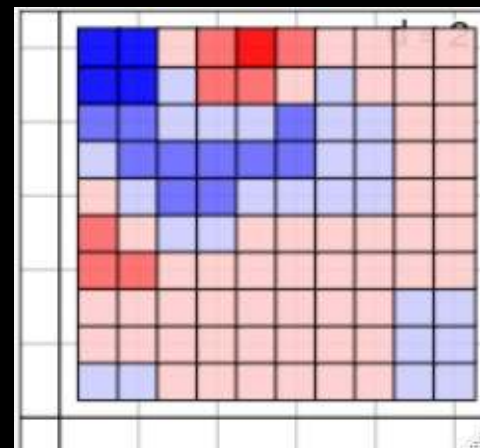


Dray et al (2012) Ecological Monographs
Jombart, Dufour & Dray (2009) Ecography

Décomposition des structures



Prédicteurs spatiaux (MEM)



Variable spatiale latente

Analyses Spatiales Multivariées

- Comment intégrer l'information spatiale dans une analyse ?
- Quelles méthodes pour faire parler les structures spatiales multivariées ?
- Quand les structures spatiales posent un problème ?

Les deux visages de l'autocorrélation



Proxies de processus non mesurés

Les deux visages de l'autocorrélation

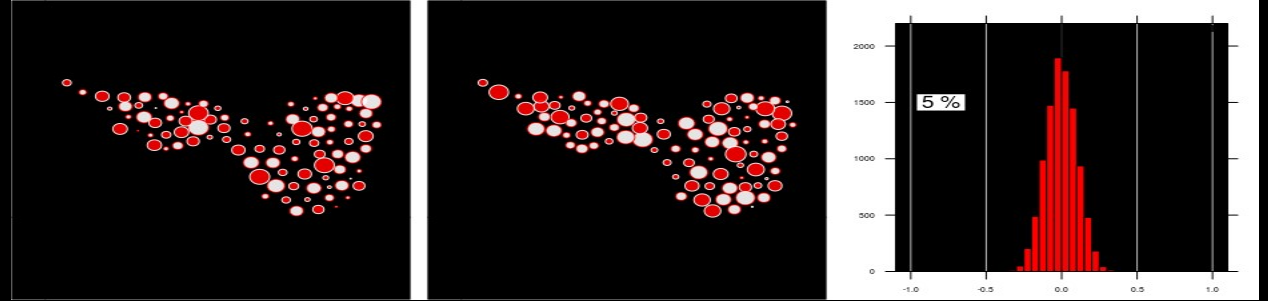


Proxies de processus non mesurés

Observations non-indépendantes

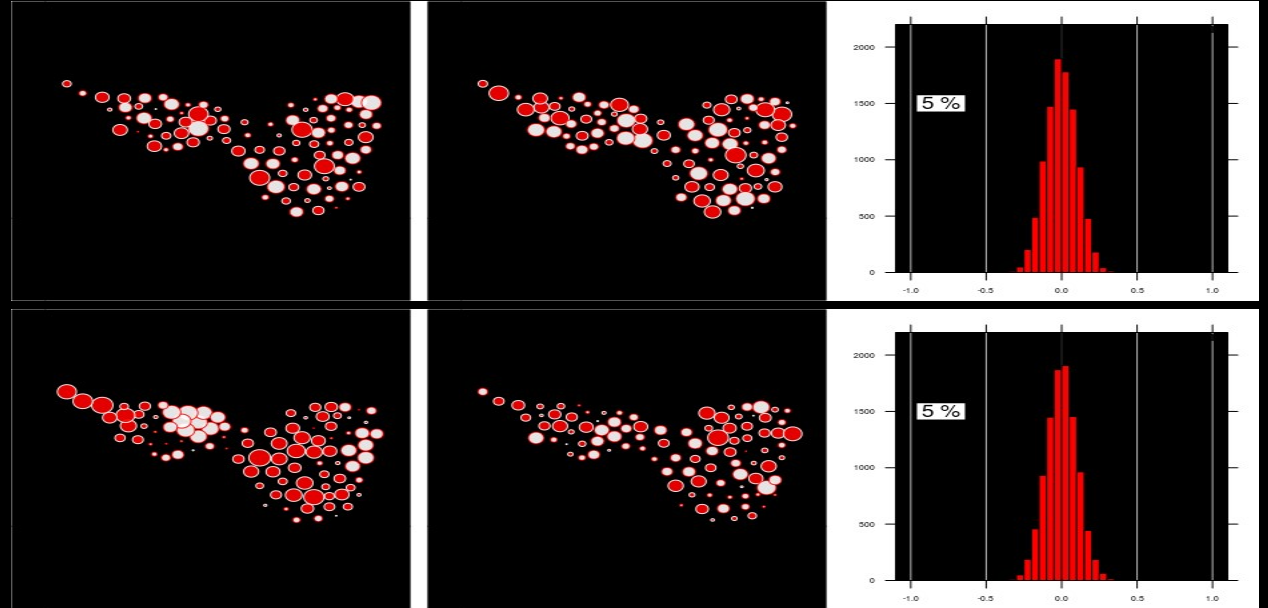
Inférence et autocorrélation spatiale

- Pas de structure



Inférence et autocorrélation spatiale

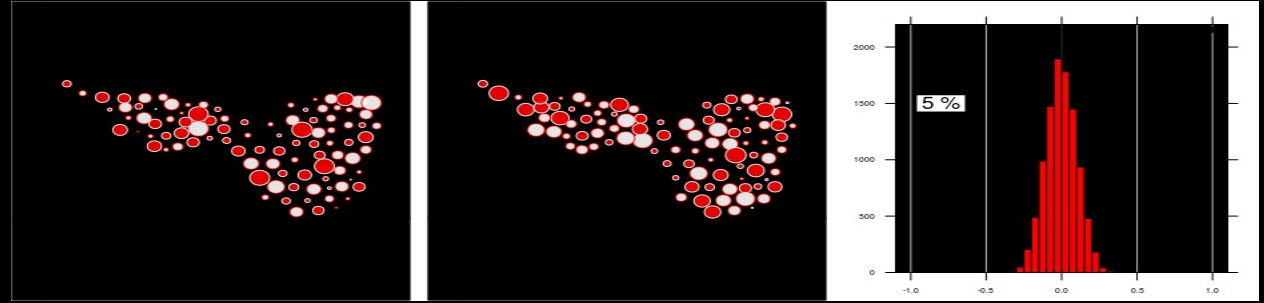
- Pas de structure



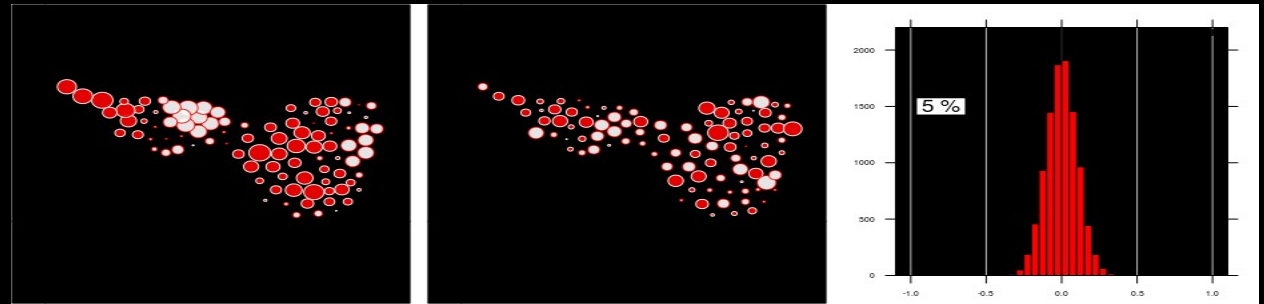
- Une structure spatiale

Inférence et autocorrélation spatiale

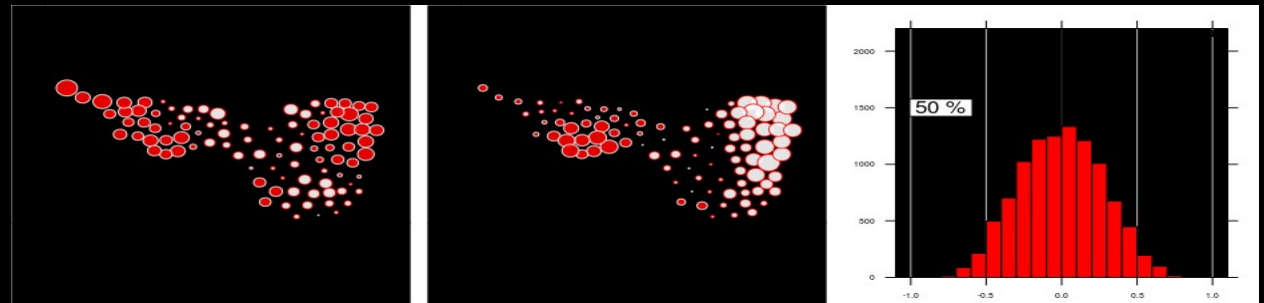
- Pas de structure



- Une structure spatiale

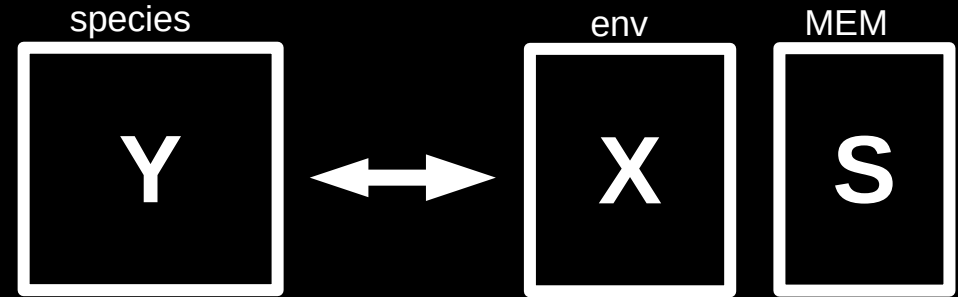


- Deux structures spatiales indépendantes



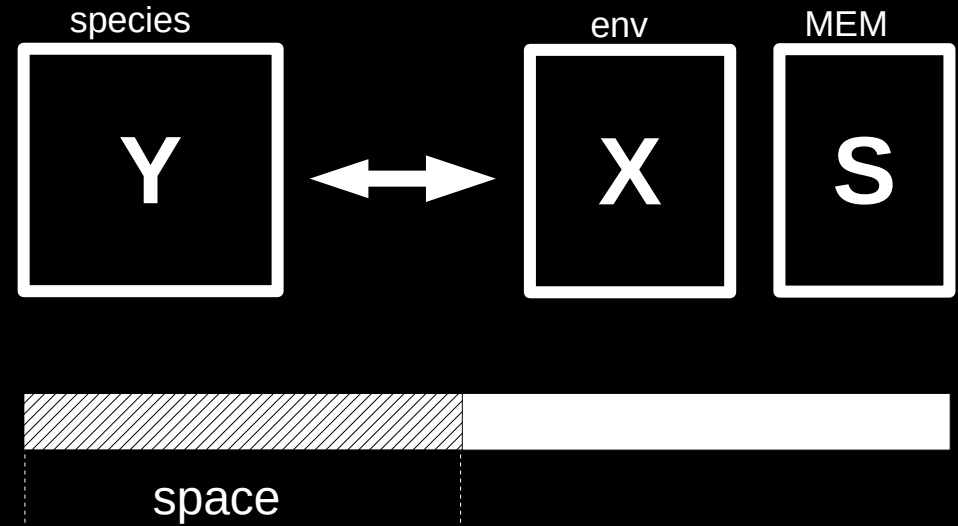
Conséquences en multivarié

- X and Y independent
X and Y spatially structured



Conséquences en multivarié

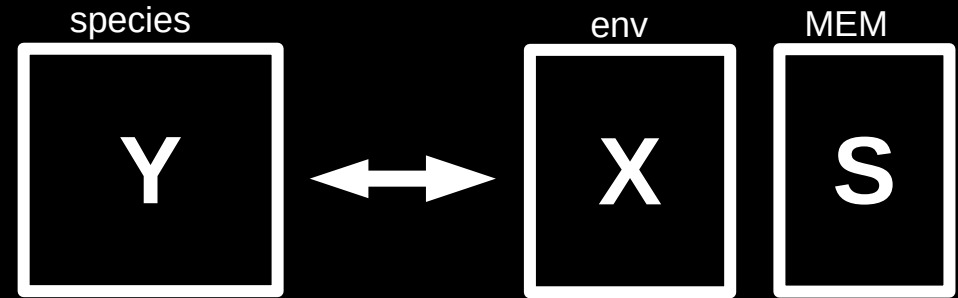
- **X** and **Y** independents
X and **Y** spatially structured



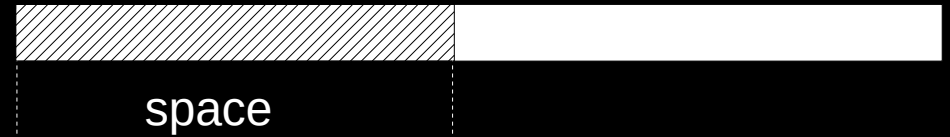
- In theory

Conséquences en multivarié

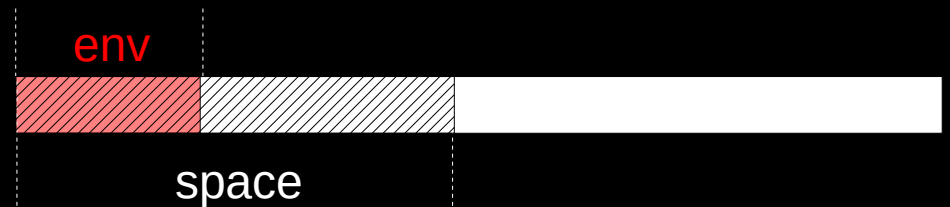
- X and Y independent
 X and Y spatially structured



- In theory

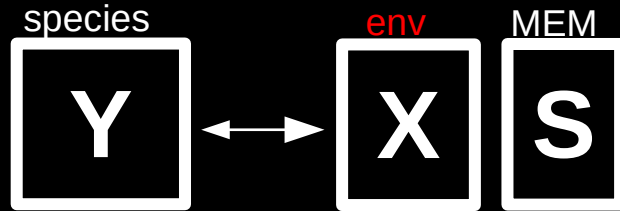


- In practice

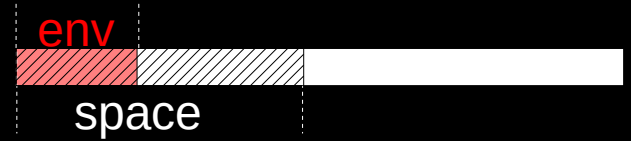


Modèles nuls contraints

- Observed data

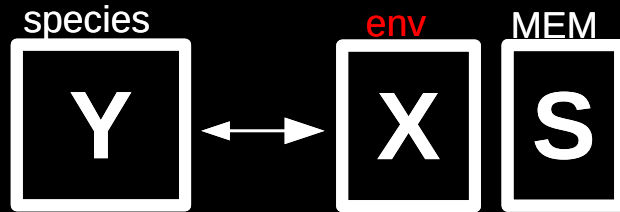


Obs :

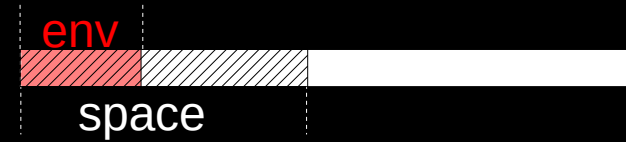


Modèles nuls contraints

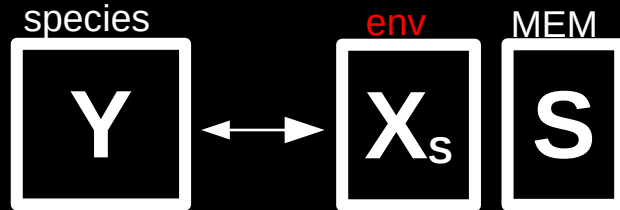
- Observed data



Obs :

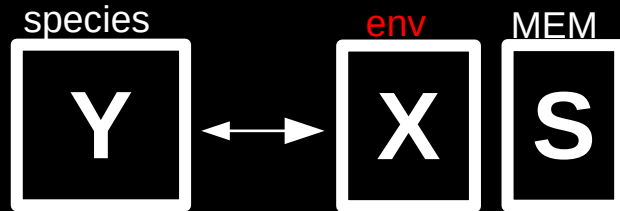


- Simulated data

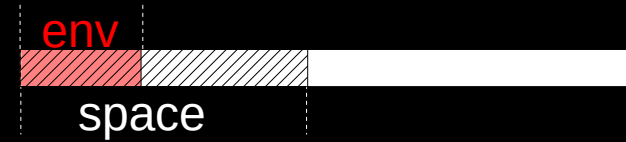


Modèles nuls contraints

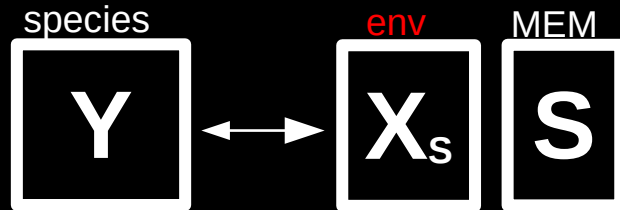
- Observed data



Obs :



- Simulated data

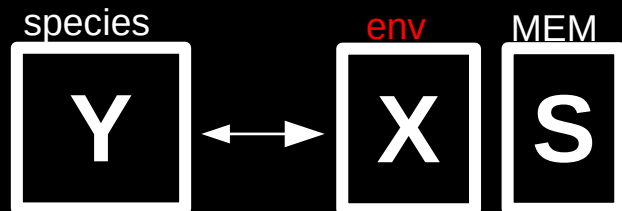


Sim 1 :

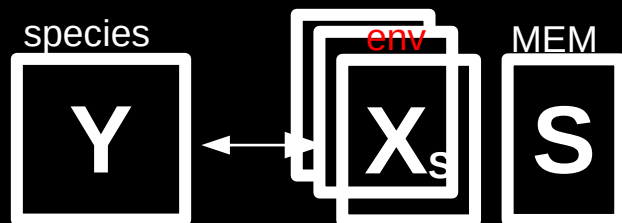


Modèles nuls contraints

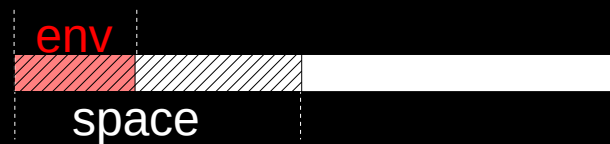
- Observed data



- Simulated data



Obs :



Sim 1 :



Sim 2 :



.

.

.

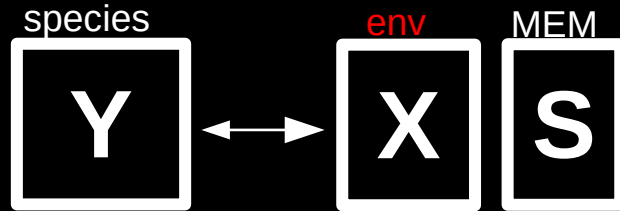
.

Sim 999 :

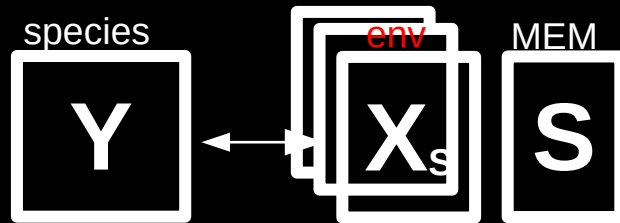


Modèles nuls contraints

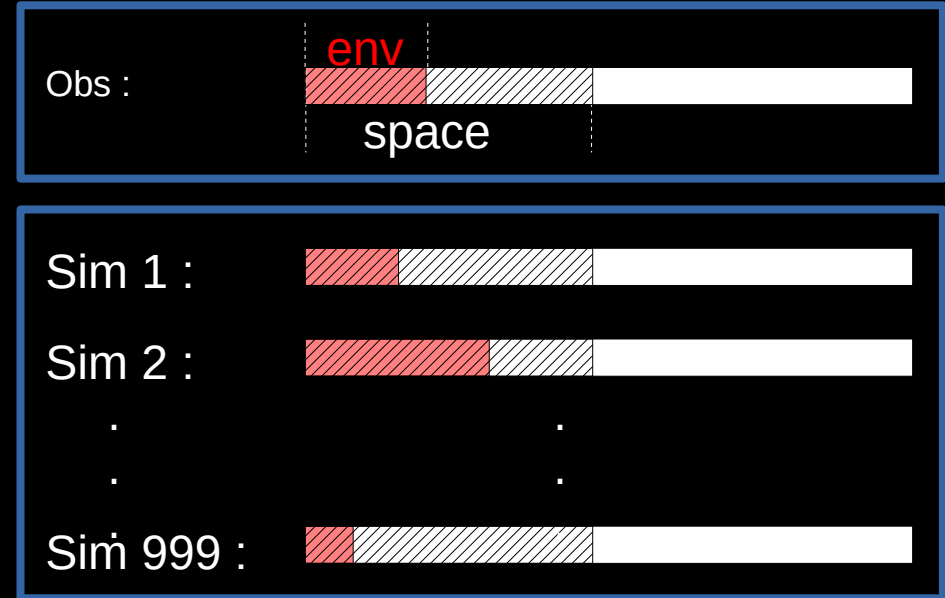
- Observed data



- Simulated data

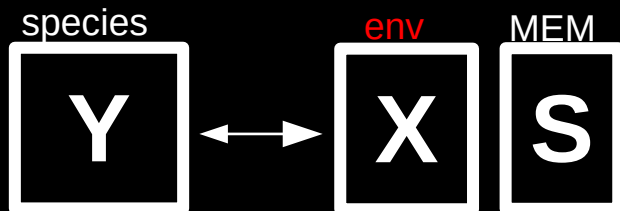


- Comparison

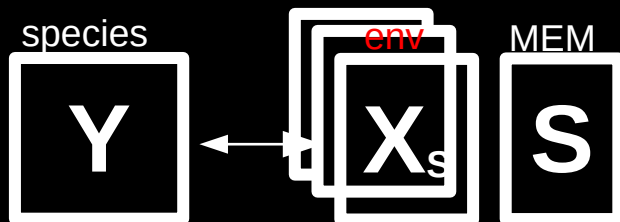


Modèles nuls contraints

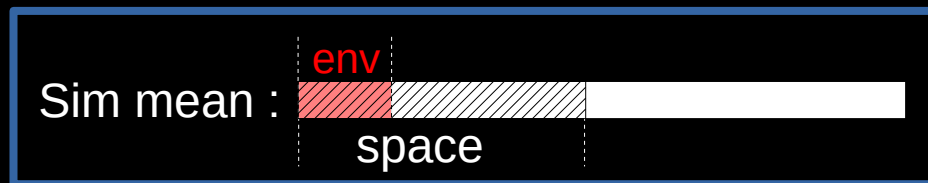
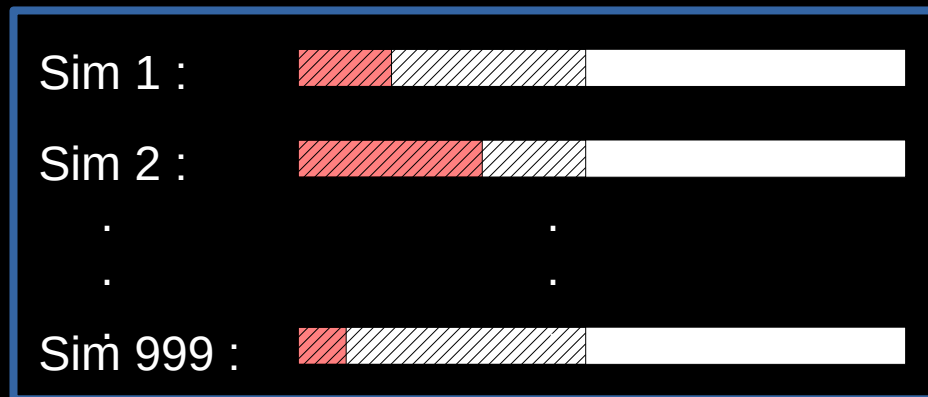
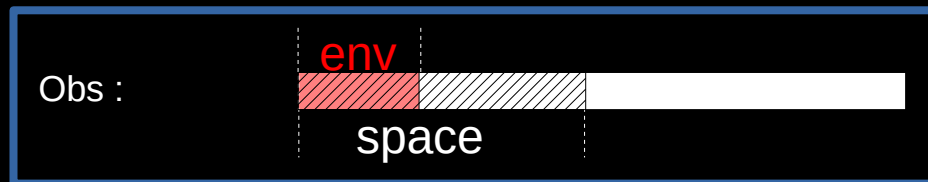
- Observed data



- Simulated data



- Comparison



Moran spectral randomization

Moran's coefficient decomposition :

$$MC(\mathbf{x}) = \sum_{i=1}^{n-1} \lambda_i \text{cor}^2(\mathbf{x}, \mathbf{v}_i)$$

Variable decomposition:

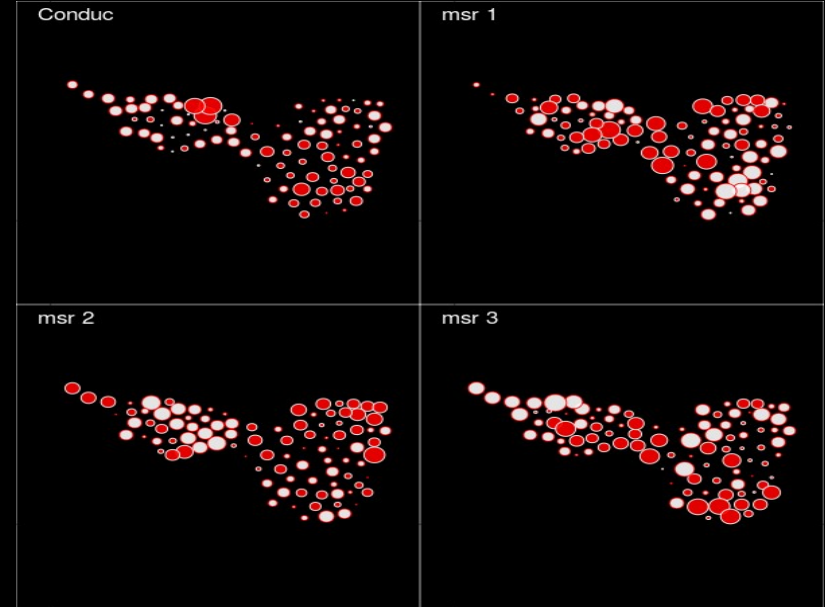
$$\mathbf{x} = \bar{x} + s(\mathbf{x}) \sum_{i=1}^{n-1} \text{cor}(\mathbf{x}, \mathbf{v}_i) \mathbf{v}_i$$

Random replicates:

$$\mathbf{x}_{sim} = \bar{x} + s(\mathbf{x}) \sum_{i=1}^{n-1} a_i \mathbf{v}_i$$

with $a_i = \pm \text{cor}(\mathbf{x}, \mathbf{v}_i)$

msr (**X**, **mymem**)



Conclusions

- Différents moyen d'intégrer l'information spatiale et multivariée
- Espace comme signature de processus non-mesurés

$$(Y \mid X) \leftrightarrow S$$

- Ou comme paramètre de nuisance en inférence

$$(Y \sim X) \mid S$$

GitHub

<https://github.com/sdray/adespatial>

Vignette

```
vignette("tutorial", package = "adespatial")
```